Transformer-Ensemble-Based Implicit Spectral–Spatial Functions for Arbitrary-Resolution Hyperspectral Pansharpening

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***Abstract*— Arbitrary-resolution hyperspectral (ARHS) pansharpening seeks to enhance hyperspectral (HS) images to any desired spatial resolutions by integrating HS images with their connected panchromatic (PAN) images, which is obviously different from standard HS pansharpening where HS data are merely pansharpened to the spatial resolutions of PANs. Therefore, ARHS pansharpening has a great potential to broaden the applications of the traditional pansharpening technique. However, ARHS pansharpening is faced with some inherent obstacles, including how to produce HS images with arbitrary resolutions beyond the training scales and how to maintain high spatial–spectral fidelities at any pansharpening scales. Implicit neural representations (INRs) is able to parametrically tune neural networks to accommodate unanalytical continuous characteristics of real-world scenes, and thus offer natural representations for ARHS pansharpening. The core of pansharpening INR design is to construct suitable implicit spectral–spatial functions (ISSFs). In this work, we propose two innovatively novel transformer-ensemble (TE)-based ISSFs, i.e., preintegrated ISSF (preISSF) and postintegrated ISSF (postISSF), which use suitable coordinate operation, novel ensemble feature extraction, and coordinate query mapping to build strong spectral–spatial reconstruction capabilities, where some new tactics such as mode-1 transformer decomposition and explicit–implicit positional encoding system are involved. We test our preISSF and postISSF on three datasets. The experimental results verify the excellent performance of our methods.**

***Index Terms*— Arbitrary-resolution hyperspectral (ARHS) pansharpening, hyperspectral (HS) images, implicit function, transformer-ensemble (TE).**

NOMENCLATURE

Symbol	Description
$\mathbf{P}$	Observed PAN image.
$\mathbf{H}$	Observed LRHS image.
$\mathbf{H}^{\uparrow}$	Preinterpolated version of the observed LRHS image.
$\widetilde{\mathbf{P}}$	Synthetic reduced-resolution PAN image.
$\widetilde{\mathbf{H}}$	Synthetic reduced-resolution LRHS image.
$\widetilde{\mathbf{H}}^{\uparrow}$	Preinterpolated version of the reduced-resolution LRHS image.
$\mathbf{P}_t$	PAN image for training phase.
$\mathbf{H}_t$	LRHS image for training phase.
$\widetilde{\mathbf{P}}_t$	Synthetic reduced-resolution PAN image for training phase.
$\widetilde{\mathbf{H}}_t$	Synthetic reduced-resolution LRHS image for training phase.
$\widetilde{\mathbf{H}}_t^{\uparrow}$	Preinterpolated version of the synthetic reduced-resolution.
$\widehat{\mathbf{H}}_a$	Pansharpened HRHS image with an arbitrary resolution
s	Pansharpening scale, i.e., ratio between the spatial resolution of $\widehat{\mathbf{H}}_a$ and that of $\mathbf{H}$.
s'	Intermediate scale, i.e., ratio between the spatial resolution of $\widehat{\mathbf{H}}_a$ and that of $\mathbf{P}$ (for standard HS pansharpening, $s' = 1$).

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I. INTRODUCTION

HYPERSPECTRAL (HS) images consist of many continuous narrow spectral bands, which offer huge potential to enhance many remote sensing tasks [1], [2], [3]. Nevertheless, HS images are usually with relatively low spatial resolutions owing to the inherent constraints between spectral and spatial resolutions of spectroscopy imaging, which leads to obvious limitations in some practical applications [4], [5], [6]. To overcome the weakness, one way is to directly improve the capabilities of imaging spectrometers, which means designing and producing advanced instruments with expensive cost. The other way is to rely on computation imaging to promote acquired HS data in a low-cost soft fashion, where HS pansharpening is a typical example. HS pansharpening cooperates HS images with their connected panchromatic (PAN) images to perform HS image enhancement [5], [7].

Almost all of the existing HS pansharpening methods are based on **standard HS pansharpening**, which aims to enhance HS images to the spatial resolutions of connected PAN images [5], [6], [8], [9], [10]. Some of those methods

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evolve from classical multispectral (MS) pansharpening, such as including component substitution (CS) [11], [12] and variational optimization (VO) [7], [11], [13], [14], [15]. However, HS images contains much more spectral bands compared with MS images, which spans apparent gaps between their spectral resolutions and between their spectral ranges. This implies that specially designed HS pansharpening methods are required [5], [8], [16]. Bayesian-based methods usually tend to obtain the posterior distribution of the target full-resolution data by means of a suitable prior of the data and the spectral demixing ability of the HS observation. Matrix factorization relies more on the dictionary and subspace assumption to attain optimal factorizations of the connected matrices. Over the past few years, convolutional neural networks (CNNs) with relatively strong feature representation ability are widely used for pansharpening [9], [17], [18], [19], [20], [21]. In HS pansharpening design, He et al. [22], [23] proposed a spectral prediction structure that aims at strengthening the spectral fidelity of the image, and further designed HSpeNets which fully utilize skip connections to reconstruct the spectral information of the final image. Zheng et al. [24] constructed a deep residual pansharpening network. Qu et al. [25] designed a CNN structure that uses cross-guided attention to extract the PAN spatial information and the HS spectral information. Zhuo et al. [21] established a deep-shallow fusion structure-based pansharpening network. He et al. [6] produced six dynamic pansharpening networks. More recently, to relief the localization limitation of CNNs, transformers are eventually introduced into HS pansharpening [26]. Bandara and Patel [27] first applied this technique to HS pansharpening, and then Shang et al. [28] extended it with an inspired model.

Being faced with diverse applications of HS images, standard HS pansharpening, which just rigidly strengthens an HS image to the fixed spatial resolution of its PAN counterpart, may not suffice to meet practical requirements [29], [30]. Therefore, arbitrary-resolution HS (ARHS) pansharpening is proposed, which is able to produce HS images with arbitrary desired spatial resolution [29], [30]. However, technically, AHRS pansharpening is much more challenging than standard HS pansharpening. First, pansharpening models are usually learned under fixed training scales, but have to infer incoming data to any possible desired scale which may beyond training scales. Second, the target spatial resolution of HS zoom may be different from PAN, which means the PAN data offer less direct compensation to support recovering HS details to that target resolution, especially at many continuous bands simultaneously. He et al. [29], [30] first designed a two-stage relay CNN to deal with ARHS pansharpening by involving offset localization-sharing across training and inferring, and then they further constructed a variable subpixel-based ARHS pansharpening thread following the consideration of spatial pattern invariance across training and inferring.

Implicit neural representations (INRs) aim to parametrically tune neural networks to estimate continuous functions [31], [32], [33], [34], [35], [36], [37]. Real-world scenes around us are intrinsically continuous, which implies their mathematical representations of continuous multivariate functions. However, those functions are usually complex and not analytical, and thus usually cannot be parameterized into mathematical formulas. In this situation, neural networks can be designed to implicitly represent those continuous functions and to implicitly encipher the continuous functions into their network parameters, where no explicit mathematical formulations are required. Therefore, INRs can be naturally associated with infinite resolution representations of natural scenes and any desired resolution images can be obtained by their samplings. In the scenario of dealing with complex observed scenes, a practical INR usually involves some specific stages. Typically, a preliminary encoding process can be employed to compact the original input and refine the prior of the observation subspace, and then an implicit function can be used to construct the INR mapping. The design of the implicit function is at the core of developing an INR, which inputs pixel coordinates and coordinate-related data and then establishes a mapping between the data and the images at desired resolutions. Some researchers were devoted to constructing implicit functions to establish effective INR mappings for the traditional super-resolution task. Chen et al. combined coordinate-based local ensemble with MLP to design a local implicit image function (LIIF) to perform super-resolution INR [35]. Lee and Jin [38] construct a local texture ensemble and thus a high frequency sensitive implicit function. Yao et al. [39] modeled texture detail distributions using normalizing flow-based scaling factors to build a local implicit normalizing flow. Cao et al. [40] and Chen et al. [41] introduced self-attention into implicit function to accommodate similarities of features in a relatively large field.

This work is dedicated to designing implicit spectral–spatial functions (ISSFs) for ARHS pansharpening. More specifically, we develop transformer-ensemble (TE)-based ISSFs for ARHS Pansharpening. The crucial consideration in our design is to strengthen ISSFs' fidelities to the coordinate-relevant data characteristic to ensure strong spectral–spatial reconstruction abilities and scale generalizations. Other considerations include how to couple HS and PAN data sources and how to accommodate their spectral–spatial characteristics. To the best of our knowledge, this is the first work exclusively focusing on designing ISSFs for the ARHS pansharpening task. More specifically, major innovative contributions of our work can be identified as follows.

1) *Preintegrated ISSF (preISSF) Scheme:* In the novel scheme, spectral–spatial contextual integration is fulfilled in ensemble feature extraction before coordinate query mapping. Our preISSF scheme uses a newly designed TE process and a newly designed coordinate query mapping process to implicitly and explicitly strengthen the coordinate-feature association, where the TE is composed of an internal subensemble and an external subensemble, which extract contextual information across the spectral–spatial latent code, the refined latent code, and the original input LRHS plus PAN.
2) *Postintegrated ISSF (postISSF) Scheme:* In the novel scheme, spectral–spatial contextual integration is accomplished after ensemble feature extraction, while in the duration of coordinate query mapping. PostISSF comprises a newly designed spatial TE (SpaTE) and a

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newly designed spectral–spatial coordinate query fusion to strengthen the coordinate-feature association, where the former contains a SpaTE and a spectral TE (SpeTE) and the latter performs a query mapping-level fusion.

3) *Mode-1 Decomposition-Based Latent Transformer:* In preISSF, we need to design a latent transformer to extract global contextual information, which incurs a high computation burden. Therefore, we design a novel mode-1 decomposition-based latent transformer by using a separable assumption of the voxel-wise attention operator. As it is fulfilled by the integration of mode-1 attention along all the single tensor modes, thus significantly reduces the computation burden.
4) *Explicit–Implicit Positional Encoding Systems:* We propose the concept of explicit–implicit positional encoding and develop novel explicit–implicit positional encoding systems for preISSF and postISSF. PreISSF embeds implicit positional encodings in the spatial/spectral transformer upsampling of internal subensemble and in that of external subensemble, and meanwhile, an explicit positional encoding in the coordinate query mapping. PostISSF sets implicit positional encodings to the SpaTE and the SpeTE, and meanwhile, two explicit positional encodings in the spectral–spatial coordinate query fusion. Specifically, we newly design a spatial steerable harmonic-based coordinate function for the explicit positional encoding, which builds a more complex coordinate manifold to help better fit high-frequency data.

The rest of our work is divided into several parts. Section II introduces our problem consideration. Section III is the design of our preISSF. Section IV describes the details of our postISSF. Section V shows our experimental results on several datasets. Section VI gives the conclusion of our work.

II. Problem Consideration

INRs are able to effectively approximate continuous functions associated with continuous real-world scenes, which implies we can develop INR-based ARHS pansharpening to adjust HS image resolutions arbitrarily, where the design of suitable ISSFs is critical. Nomenclature gives the mathematical notation adopted in later descriptions.

A. ARHS Pansharpening INR

ARHS pansharpening is usually a learning-based process. If introducing INRs into ARHS pansharpening, we have the training phase and the inferring phase formulated as

$$\boldsymbol{\theta}^{\text{opt}} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}\left(\Phi(\boldsymbol{c}_t; \boldsymbol{r}_t = \Psi\left(\widetilde{\mathbf{H}}_t^{\uparrow}, \widetilde{\mathbf{P}}_t\right), \boldsymbol{\theta}), \mathbf{H}_t\right) \tag{1}$$

and

$$\widehat{\mathbf{H}}_a = \Phi\left(\boldsymbol{c}_e; \boldsymbol{r}_e = \Psi\left(\mathbf{H}^{\uparrow}, \mathbf{P}\right), \boldsymbol{\theta}^{\text{opt}}\right) \tag{2}$$

respectively, where (1) denotes the training phase and (2) stands for the inferring phase, Φ denotes the pansharpening network and $\boldsymbol{\theta}$ represents its parameters during the learning, $\boldsymbol{\theta}^{\text{opt}}$ denotes the training result, $\mathcal{L}$ is the loss of the pansharpening network, $\boldsymbol{c}_t = (x, y, b)$ denotes the spectral–spatial coordinates of the training data, $\boldsymbol{r}_t$ represents priors connected to training data, $\boldsymbol{c}_e = (x_e, y_e, b_e)$ stands for any desired samplings of continuous spectral–spatial coordinates, $\boldsymbol{r}_e$ represents priors connected to inferring data, and Ψ represents the operation of extracting priors. In (1) and (2), the network Φ is in fact the ISSF to approximate continuous spectral–spatial functions, whose parameters $\boldsymbol{\theta}$ are able to encipher those continuous spectral–spatial functions. Once the parameters $\boldsymbol{\theta}^{\text{opt}}$ are learned in the training phase, the rule to yield the continuous spectral–spatial functions for given priors is determined.

B. Implicit Spectral–Spatial Function

ISSF is the core of an effective ARHS pansharpening INR. The major consideration for our ISSF design is to strengthen the association of coordinates and spectral–spatial features to produce strong spectral–spatial reconstruction abilities and scale generalizations. Suppose $\mathbf{g}$ represents a continuous spectral–spatial scene in the real-world, which is a 3-D continuous cube, and the scene is observed with an LRHS image and its connected PAN image. Let the latent code $\boldsymbol{r}$ encode the prior information of the LRHS and the PAN. An ISSF can be further formulated as

$$\begin{aligned}\mathbf{g} &= \Phi\left(\boldsymbol{c}; \boldsymbol{r}, \widetilde{\mathbf{H}}^{\uparrow}, \widetilde{\mathbf{P}}, \boldsymbol{\theta}\right) \\ &= \Phi_q\left(\boldsymbol{c}; \mathbf{M} = \Phi_m\left(\boldsymbol{r}, \widetilde{\mathbf{H}}^{\uparrow}, \widetilde{\mathbf{P}}; \boldsymbol{\theta}_m\right), \boldsymbol{\theta}_q\right)\end{aligned} \tag{3}$$

where $\boldsymbol{c}$ denotes spectral–spatial coordinates, $\boldsymbol{r}$ stands for the associated data prior, $\Phi_m(\cdot)$ denotes the ensemble to ensure the fidelity of its features to the spectral–spatial coordinates $\boldsymbol{c}$, $\boldsymbol{\theta}_m$ denotes the parameters of the ensemble, $\Phi_q(\cdot)$ represents the spectral–spatial query function, and $\boldsymbol{\theta}_q$ stands for the parameters of the query. That is, an ISSF is composed of an ensemble and a query mapping, and therefore $\boldsymbol{\theta} = [\boldsymbol{\theta}_m, \boldsymbol{\theta}_q]$ comprises the parameters of the ISSF Φ.

Considering LRHS data and the connected HRHS data are sampled from the same continuous spectral–spatial scene, the related low-resolution coordinates and high-resolution coordinates can be unified with the following ISSF representation:

$$\mathbf{g} = \Phi_q\left(\left(\frac{x}{s}, \frac{y}{s}, b\right); \mathbf{M} = \Phi_m\left(\boldsymbol{r}, \widetilde{\mathbf{H}}^{\uparrow}, \widetilde{\mathbf{P}}; \boldsymbol{\theta}_m\right), \boldsymbol{\theta}_q\right). \tag{4}$$

The unified low-resolution coordinates and high-resolution coordinates can be further normalized to $[0, 1]$ to accomplish the unified coordinate normalization, which build coordinate-sharing across the training and the inferring.

In the following, we develop two TE-based ISSFs, i.e., preISSF and postISSF, following the above coordinate normalization.

III. Preintegrated ISSF

Pansharpening INRs offer possibilities to derive HS data at any desired resolutions from the observed LRHS images and PAN images, where suitable ISSFs should be designed to directly decipher spectral–spatial information to yield data in arbitrary target resolutions. The design of an ISSF should include two major considerations, i.e., ensemble strategy and query mapping strategy.

Fig. 1 graphically illustrates the structure of our innovative preISSF. In preISSF, spectral–spatial information integration

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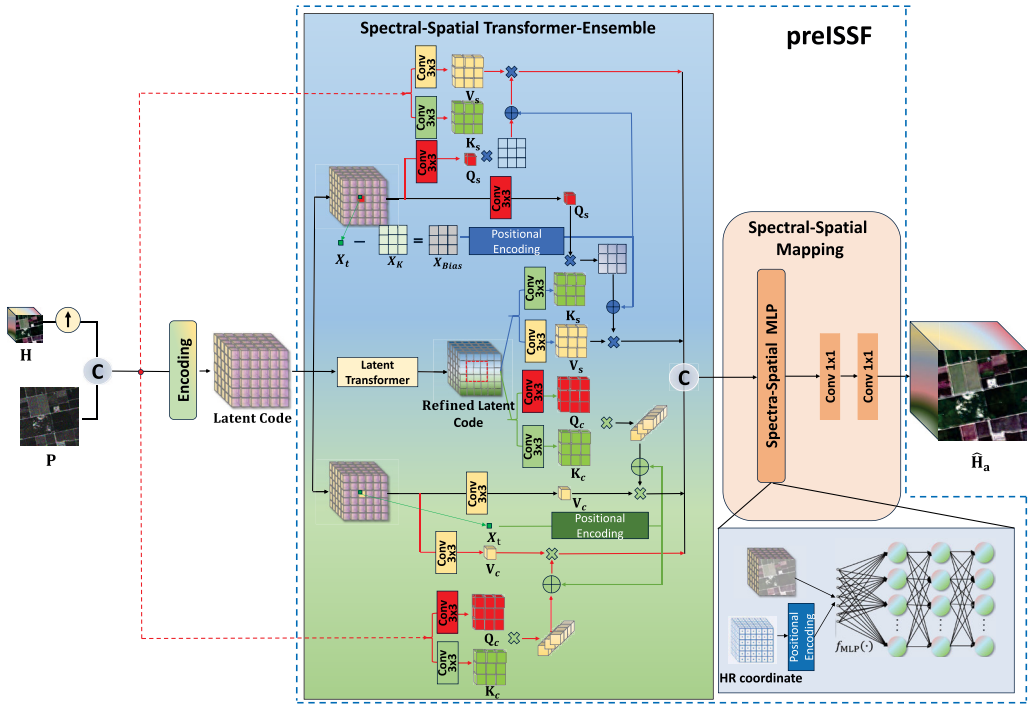


Fig. 1. Overall architecture of preISSF. Spectral–SpaTE produces ensemble features of global spectral–spatial context from the original data and the latent code, and coordinate query mapping decodes the ensemble features to generate ARHS image.

is completed during extracting ensemble features, which is before coordinate query mapping. As shown in the figure, a prepansharpening module is used as the encoding subnetwork to produce the latent code, and then preISSF is connected. Our preISSF mainly comprises a spectral–SpaTE and a spectral–spatial query mapping. The ensemble seeks to extract ensemble features for spatial coordinates in the continuous HS image domain, which are beneficial for the succeeding target resolution recovery. The query mapping aims at using the features above to estimate the continuous function that maps the querying coordinates to the spectral–spatial data at any desired resolutions.

A. Spectral–SpaTE

Our preSSIF exploits the spectral–SpaTE to perform global and local attentions on both latent code and PAN/HS data. That TE produces highly effective spectral–spatial features for spatial coordinates in the continuous HS image domain. Those ensemble features are expected to improve deciphering HS image at any desired arbitrary resolutions. Our TE is mainly composed of an internal subensemble and an external subensemble.

1) Internal Subensemble: The internal subensemble is designed to yield ensemble features from the latent code. It includes a latent transformer and a cross-latent code attention-based upsampling, where the former is used to extract global context information and the latter is for upsampling by handling local context cross the original latent code and the refined latent code. Since spectral–spatial context is contained in the 3-D data cube, intuitively, the latent transformer should be a module in a 3-mode tensor representation. However, this incurs computation difficulty. In the following, we tend to design a mode-1 decomposition-based method to deal with the 3-mode self-attention computation.

a) Mode-1 decomposition of latent transformer: For the latent transformer, we calculate the query Q, the key K, and the value V on the latent code, respectively. Assume the latent code L is with the size of $H \times W \times B$, we can build the transformer in the 3-mode tensor representation as follows:

$$\begin{aligned} Q &= f(L), \quad K = f(L), \; V = f(L) \\ Q_v &= \text{vec}(Q), \quad K_v = \text{vec}(K), \; V_v = \text{vec}(V) \\ S &= \text{Softmax}\left(\frac{Q_v K_v^T}{\sqrt{C}}\right) \\ M_v &= V_v \times S \\ M &= \text{ten}(M_v) \end{aligned} \tag{5}$$

where $f(\cdot)$ denotes the linear layer, $\text{vec}(\cdot)$ represents the vectorization unfolding of a 3-mode tensor, $S \in R^{HWB \times HWB}$ denotes the similarity matrix, $\text{ten}(\cdot)$ stands for the tensorization of a vector, and $M \in R^{H \times W \times B}$ represents the result of spectral–spatial attention modulation, which is a 3-mode tensor with the same size as the latent code L. When performing the self-attention mechanism, similarity calculations are conducted in a pixel-versus-three mode tensor, which would lead to a high computation burden. The computational complexity of the above 3-D transformer is $O(H^2W^2B^2 + HWB^2)$.

To reduce the computation burden of the original latent transformer, we design a mode-1 decomposition-based method. The operation in (5) can be represented with an operator T_{hwb}. Apparently, T_{hwb} is a voxel-wise attention operator, which maps an input 3-mode tensor to the connected 3-D-attention modulated tensor with the same size. Assume the attention operator T_{hwb} is separable along three modes of a tensor, i.e., the related 3-D-attention can be approximated by

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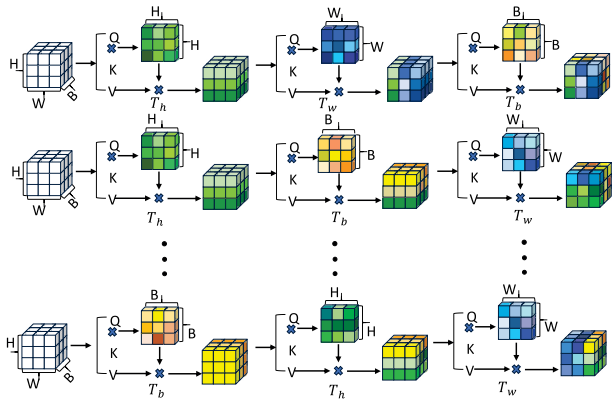


Fig. 2. Structure of our mode-1 decomposition-based latent transformer.

the integration of mode-1 attention along all the single mode of the processed tensors. Then, we have

$$T_{hwb} = T_h T_w T_b + T_h T_b T_w + T_w T_h T_b + T_w T_b T_h + T_b T_w T_h + T_b T_h T_w \tag{6}$$

where T_h, T_w, and T_b represent mode-1 attention operators along the first, the second, and the third mode, respectively. Taking T_h for example, it is of the form

$$\begin{aligned} Q^{H\times WB} &= f(D),\ K^{H\times WB} = f(D),\ V^{H\times WB} = f(D) \\ S^{H\times H} &= \text{Softmax}\left(\frac{QK^T}{\sqrt{C}}\right) \\ M_h &= V \times S. \end{aligned} \tag{7}$$

T_w and T_b perform similar mode-1 attention operation as above (7), which can achieve similarity matrices $S^{W\times W}$ and $S^{B\times B}$, respectively.

When we need to apply 3-D attention T_{hwb} on latent code L, it can also be represented by mode-1 attention operators as follows:

$$T_{hwb}L = T_h T_w T_b L + T_h T_b T_w L + T_w T_h T_b L + T_w T_b T_h L + T_b T_w T_h L + T_b T_h T_w L. \tag{8}$$

Equation (8) reveals that the 3-mode latent transformer can be alternatively performed with superposition of six subtransformers (as shown in Fig. 2). Each of the subtransformers is constituted by serializing three mode-1 transformers, which are computed along the first, the second, and the third modes, respectively. Such a computation exhausts all the combinations of mode-1 transformers along three modes, and therefore can be viewed as the approximation of the original 3-mode tensor-based transformer. Compared with the original 3-mode latent transformer, our mode-1 decomposition-based latent transformer significantly reduces the computation burden since it only requires calculating the similarity between pixels along one dimension of the 3-mode tensor. The computational complexity of the mode-1 attention operator along the first mode is $O(H^2WB)$. Therefore, the computational complexity of our mode-1 decomposition-based latent transformer is $O(H^2WB + W^2HB + B^2HW)$.

After latent transformer processing on the latent code, we have refined the latent code, which contains global spectral–spatial context information.

b) Cross-latent code attention-based upsampling: To further process 3-mode context information extracted from the latent transformer for the continuous HS images domain, we design a cross-latent code attention-based upsampling. The upsampling comprises two specific transformer-based upsampling, one is a spatial transformer-based and the other is a spectral transformer-based. The former is used to construct spatial cross-context for coordinates in the continuous HS domain, while the latter is used to utilize spectral cross-context. Both transformer-based upsamplings use a cross-latent mechanism to deal with spatial/spectral local context. The cross latent mechanism is built between the original latent code and the refined latent code.

Spatial transformer upsampling uses the cross-latent mechanism to deal with spatial local context. As shown in Fig. 1, we calculate Q_s from the latent code to approximate pixels at a target resolution, which is used as a query for the spatial transformer. Then, we generate K_s and V_s from the refined latent code in order to utilize the preextracted 3-D context information. Q_s, K_s, and V_s are utilized to form spatial transformer-based upsampling. Fig. 3 graphically illustrates the detail of the spatial transformer-based upsampling process. The latent code is first upsampled by nearest neighboring to obtain Q_s, where x_t denotes the continuous coordinate of the target pixel to be restored in the desired spatial resolution, and x_q is the continuous coordinate of the low-resolution pixel related to the target high-resolution pixel. K_s and V_s represent the low-resolution pixels in the ensemble area of the refined latent code. Then, spatial transformer-based upsampling for each pixel of Q_s is performed. Q_s and K_s are first used to implement dot product to produce a spatial weight map, which reflects similarities between the target pixel x_t of latent code and its surrounding pixels in the related ensemble area of refined latent code. Thus, the spatial transformer upsampling can be formulated as

$$\begin{aligned} N &= T_{hwb}L \\ Q_s &= f_{Qs}(L), \quad K_s^{(i,j)} = f_{Ks}(N), \quad V_s^{(i,j)} = f_{Vs}(N) \\ \delta_s &= \mathcal{P}\left(x_t - x^{(i,j)}\right) \\ W_s &= \text{Softmax}\left(\frac{Q_s K_s^{(i,j)}}{\sqrt{C}}\right) \times \delta_s \end{aligned} \tag{9}$$

where L stands for the latent code, C is the channel number of the embedding Q_s, $\mathcal{P}(x) = [\sin(2^0 x), \cos(2^0 x), \ldots, \sin(2^{l-1}x), \cos(2^{l-1}x)]$ is a coordinate function, f_{Qs}, f_{Ks}, and f_{Vs} are convolution layers to yield Q_s, K_s, and V_s, respectively, T_{hwb} represents the mode-1 latent transformer, and W_s represents the spatial weight map. W_s is then multiplied with V_s, that is, the pixel features in V_s are weighted one by one through the spatial weight map, which can achieve spatial ensemble in the nonlocal spectral–spatial information. The process can be formulated as

$$I_s = W_s \times V_s^{(i,j)}. \tag{10}$$

Through the operation in (10), we have the ensemble features produced by the spatial transformer upsampling.

It is worth to note that here we add a positional encoding to strengthen the spatial weight map to significantly distinguish

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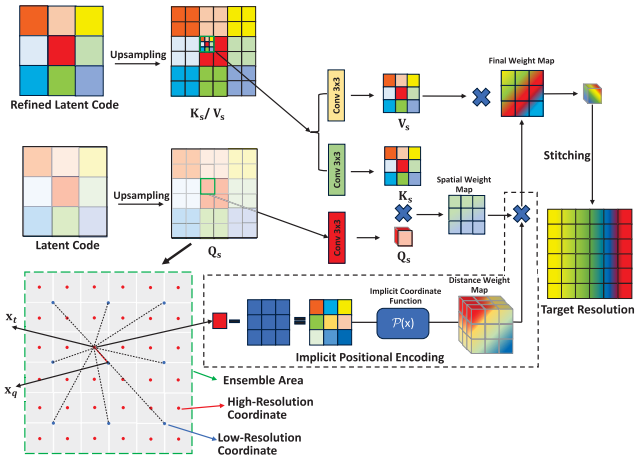


Fig. 3. Spatial transformer upsampling.

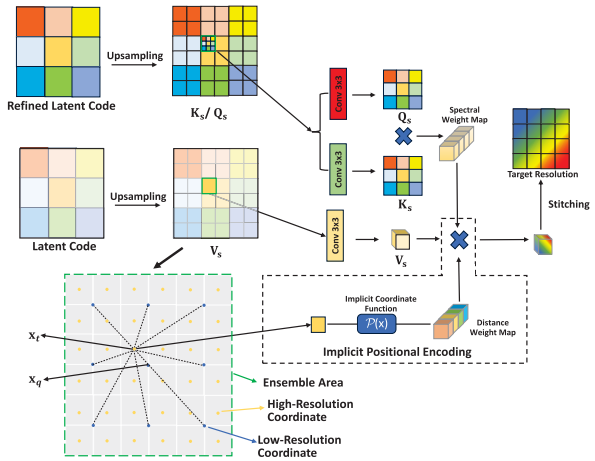


Fig. 4. Spectral transformer upsampling.

the neighboring high-resolution pixels associated with a low-resolution pixel. As shown in Fig. 3, the red pixel of latent code is associated with the same Q_s, K_s, and V_s, which means that they achieve similar results in the spatial weight map. In our positional encoding, we use the continuous coordinate of the target high-resolution pixel and the continuous coordinates of its surrounding low-resolution pixels to compute the Euclidean distances between the target pixel and the its surrounding pixels. Then, the distances go through the coordinate function $\mathcal{P}(x)$ to produce a distance weight map. Then, the distance weight map is multiplied with the spatial weight map to yield the final weight map W_s, which is used to modulated V_s to yield features I_s. As depicted above, the distance information in the target resolution is implicitly encoded in the features of the ensemble, and thus we call the positional encoding process implicit positional encoding, which is in contrast to the case in later query mapping design where position information are explicitly connected with ensemble features. Correspondingly, the function $\mathcal{P}(x)$ is called an implicit coordinate function.

Spectral transformer upsampling is designed to use spectral correlation to strengthen the spectral fidelity. This transformer is also based on a cross-latent mechanism between the latent code and the refined latent code. In more detail, as shown in Fig. 4, the latent code is upsampled to generate V_c, where x_t stands for the target high-resolution pixel and the x_v denotes the low-resolution pixel related to the target pixel. Q_c and K_c are generated from the low-resolution pixels in the ensemble area of the refined latent code, which is similar to the selection of K_s and V_s in the spatial transformer upsampling case. Then, Q_c and K_c are vectorized and fulfill the dot product to generate a spectral weight map, which represents the spectral similarity of surrounding areas in the refined latent code. Thus, the connected formulation is of the form

$$\begin{aligned} &V_c = f_{Vc}(L), \quad K_c^{(i,j)} = f_{Kc}(N), \quad Q_c^{(i,j)} = f_{Qc}(N) \\ &\delta_c = \mathcal{P}(x_t) \\ &W_c = \text{Softmax}\left(\frac{Q_c K_c^{(i,j)}}{\sqrt{C}}\right) \times \delta_c \end{aligned} \tag{11}$$

where f_{Qc}, f_{Kc}, and f_{Vc} are convolution layers to yield Q_c, K_c, and V_c, respectively, and W_c represents the spectral weight map. Then, the value V_c is spectrally modulated by the spectral weight map, which achieves the spectral ensemble features from the nonlocal spectral–spatial information and can be formulated as

$$I_c = W_c \times V_c. \tag{12}$$

Similar to the spatial transformer upsampling, the spectral transformer upsampling also uses an implicit positional encoding to strengthen its spectral weight map (as shown in Fig. 4); but this implicit positional encoding directly put the high-resolution coordinate of the target pixel through the implicit coordinate function $\mathcal{P}(x)$ to yield the distance weight map to tune the spectral weight map.

2) External Subensemble: Complementary to the internal subensemble which is mainly for extracting global and local spectral–spatial context information from the latent code and the refined latent code, we additionally design an external subensemble to extract spectral–spatial context information across the LRHS/PAN data and the latent code. Different from the internal subensemble, the external subensemble calculate the spatial quantities K_s' and V_s' and the spectral quantities Q_c' and K_c' from the concatenation of HS and PAN (HPC), where the dot production of Q_S' and K_s' presents spatial similarity S_E between the target pixel x_t and the surrounding areas in HPC and the multiplication of Q_c' and K_c' gives the spectral correlation C_E of HPC. The process can be formulated as

$$\begin{aligned} &\text{HPC} = \text{Cat}(\mathbf{H}^{\uparrow}, \mathbf{P}) \\ &Q_s' = f_{Qs}'(L), \quad K_s'^{(i,j)} = f_{Ks}'(\text{HPC}), \quad V_s'^{(i,j)} = f_{Vs}'(\text{HPC}) \\ &\delta_s = \mathcal{P}\left(x_t - x^{(i,j)}\right) \\ &S_E = \text{Softmax}\left(\frac{Q_s' K_s'^{(i,j)}}{\sqrt{C}}\right) \times \delta_s \\ &V_c' = f_{Vc}'(L), K_c'^{(i,j)} = f_{Kc}'(\text{HPC}), \quad Q_c'^{(i,j)} = f_{Qc}'(\text{HPC}) \\ &\delta_c = \mathcal{P}(x_t) \\ &C_E = \text{Softmax}\left(\frac{Q_c' K_c'^{(i,j)}}{\sqrt{C}}\right) \times \delta_c \end{aligned} \tag{13}$$

where S_E and C_E stand for the spatial similarity matrix and spectral correlation matrix, respectively, f_{Qs}', f_{Ks}', f_{Vs}', f_{Qc}',

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f'_{Kc}, and f'_{Vc} are convolutional layers. Then, V'_s is modulated by S_E and V'_c is modulated by C_E to perform upsampling to yield the external spatial ensemble features E_s and the external spectral ensemble features E_c, which can be computed as

$$E_s = S_E \times V_s'^{(i,j)}$$
$$E_c = C_E \times V'_c. \tag{14}$$

Finally, the two internal ensembles and the two external ensembles are concatenated along the spectral dimension to yield the spectral–SpaTE

$$\mathrm{TE} = \mathrm{Concat}(I_s, I_c, E_s, E_c) \tag{15}$$

which is used as the ensemble features for the succeeding coordinate query mapping.

It is noteworthy that the external subensemble is also composed of a spatial transformer upsampling and a spectral transformer upsampling, which share implicit positional encodings with those two upsamplings of the internal subensemble (as shown in Fig. 1).

B. Coordinate Query Mapping

The above TE has achieved ensemble features, and then we need to design a further query mapping to infer from these features to have the final arbitrary-resolution HRHS images. More specifically, the query mapping seeks to decode the ensemble features with the given coordinates, where it can build a spectral–spatial continuous representation to restore the final HRHS data. To decode ensemble features, in the following, we design an explicit positional encoding process with a novel coordinate function, and build a concept of explicit–implicit positional encoding system to accommodate all the positional encoding tactics in our preISSF. Then, the complete design of our query mapping is given.

1) Explicit–Implicit Positional Encoding System: In the coordinate query mapping, we input spectral–spatial coordinates with the target resolution, which collaborate with ensemble features to achieve final HRHS data. Here, the information about positions is explicitly used for inferring, which contrasts with the implicit positional encoding used in TE, where the position information is hidden in extracted features (as depicted in Section III). Thus, we call the utilization of position information in the coordinate query mapping as explicit positional encoding.

When applying position information in the neural network design, many existing literatures use Sinusoidal harmonics-based coordinate function $\mathcal{P}(x) = [\sin(2^0 x), \cos(2^0 x), \ldots, \sin(2^{l-1} x), \cos(2^{l-1} x)]$ to tune positions, so as to alleviate the bias of deep networks to excessively learn low frequency components [42], [43], which has been adopted in our design of implicit positional encoding in Section III-A. Taking into consideration that using high-frequency functions to map the input of a neural network to a more complex manifold helps better fitting high-frequency data, we design a new coordinate function for our explicit positional encoding,

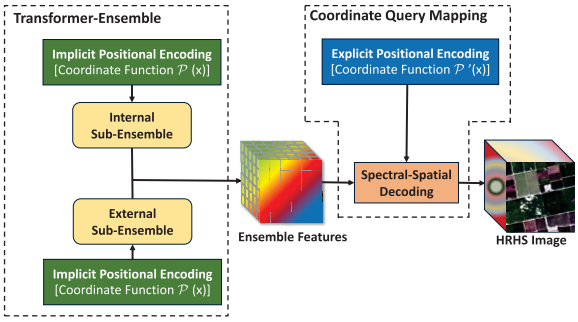


Fig. 5. Explicit–implicit positional encoding system of preISSF.

which is of the form

$$\begin{aligned}\mathcal{P}'(x) = \Big[&\ldots, g_{\sin,i}^{(x)(\theta_j)} \cdot g_{\cos,i}^{(y)(\theta_j)}, g_{\cos,i}^{(x)(\theta_j)} \cdot g_{\sin,i}^{(y)(\theta_j)}, \\ &g_{\cos,i}^{(x)(\theta_j)} \cdot g_{\cos,i}^{(y)(\theta_j)}, g_{\sin,i}^{(x)(\theta_j)} \cdot g_{\sin,i}^{(y)(\theta_j)}, \ldots, \\ &g_{\sin,\mathrm{L}-1}^{(x)(\theta_j)} \cdot g_{\cos,\mathrm{L}-1}^{(y)(\theta_j)}, g_{\cos,\mathrm{L}-1}^{(x)(\theta_j)} \cdot g_{\sin,\mathrm{L}-1}^{(y)(\theta_j)}, \\ &g_{\cos,\mathrm{L}-1}^{(x)(\theta_j)} \cdot g_{\cos,\mathrm{L}-1}^{(y)(\theta_j)}, g_{\sin,\mathrm{L}-1}^{(x)(\theta_j)} \cdot g_{\sin,\mathrm{L}-1}^{(y)(\theta_j)}\Big]\end{aligned} \tag{16}$$

where

$$\begin{aligned}g_{\cos,i}^{(x)(\theta_j)} &= \frac{1}{(2\pi)^{\frac{1}{2}}\sigma_x} \exp\left(-\frac{x^2}{2\sigma_x^2}\right) \cos\left(x\omega_{x,i}^{(\theta_j)}\right) \\ g_{\sin,i}^{(x)(\theta_j)} &= \frac{1}{(2\pi)^{\frac{1}{2}}\sigma_x} \exp\left(-\frac{x^2}{2\sigma_x^2}\right) \sin\left(x\omega_{x,i}^{(\theta_j)}\right) \\ g_{\cos,i}^{(y)(\theta_j)} &= \frac{1}{(2\pi)^{\frac{1}{2}}\sigma_y} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \cos\left(y\omega_{y,i}^{(\theta_j)}\right) \\ g_{\sin,i}^{(x)(\theta_j)} &= \frac{1}{(2\pi)^{\frac{1}{2}}\sigma_y} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \sin\left(y\omega_{y,i}^{(\theta_j)}\right).\end{aligned} \tag{17}$$

$\omega_{x,i}^{(\theta_j)} = |\omega_i| \cos\theta_j$, $\omega_{y,i}^{(\theta_j)} = |\omega_i| \sin\theta_j$, $|\omega_i| = [(\omega_{x,i}^{(\theta_j)})^2 + (\omega_{y,i}^{(\theta_j)})^2]^{1/2} = 2^i\pi$ is the magnitude of the frequency ω_i, θ_j is the angle between the horizonal direction and ω. By using various frequencies ωs and angles θs, we can generalize encoded positional information associated with spectral–spatial coordinates, which can be formulated as

$$\boldsymbol{c}_{\mathrm{pe}} = \mathcal{P}'(\boldsymbol{c}_{\mathrm{ss}}) \tag{18}$$

where $\boldsymbol{c}_{\mathrm{ss}}$ stands for the spectral–spatial high resolution coordinates and $\boldsymbol{c}_{\mathrm{pe}}$ presents the positional encoded information. Compared to the traditional coordinate function $\mathcal{P}(x)$, our newly designed coordinate function $\mathcal{P}'(x)$, which is connected to a spatial steerable filtering [44], [45], is able to achieve benefits in twofold. On the one hand, it tends to embed the positions to a more complex manifold; on the other hand, it maps those positions into a higher dimensional space. Therefore, the new spatial steerable harmonic-based coordinate function $\mathcal{P}'(x)$ is expected to be more effective for the coordinate query mapping.

Fig. 5 shows our explicit–implicit positional encoding system in preISSF, which is expected to help build high coupling of data and the connected positions. It consists of not only the explicit positional encoding based on the coordinate function $\mathcal{P}'(x)$, but also the implicit positional encoding based on the coordinate function $\mathcal{P}(x)$.

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2) Spectral–Spatial Decoding: As shown in Fig. 1, our coordinate query mapping inputs the ensemble features and the corresponding target resolution coordinates. Explicit positional encoding is then employed to fulfill the coordinate manifold transform. Finally, a spectral–spatial MLP operation maps ensemble features and the coordinate manifold to a continuous spectral–spatial representation to produce HRHS data, where two 1×1 convolutional layers are used to strengthen the spectral information. Such a spectral–spatial decoding process generates the final HS images with any desired resolutions. The spectral–spatial decoding can be represented as

$$\begin{aligned} M_O &= \mathcal{M}(\mathrm{TE}, \boldsymbol{c}_{\mathrm{pe}}) \\ F_O &= \mathcal{F}(\mathcal{F}(M_O)) \end{aligned} \tag{19}$$

where $\mathcal{M}(\cdot)$ denotes the spectral–spatial MLP and $\mathcal{F}(\cdot)$ denotes the convolutional layer with the kernel size of 1×1.

The implementation of our preISSF is summarized in Algorithm 1.

Algorithm 1 preISSF

Input: $\widetilde{\mathbf{H}}_t^{\uparrow}$, $\widetilde{\mathbf{P}}_t$, $\mathbf{H}_t$, $\mathbf{H}^{\uparrow}$, $\mathbf{P}$, inferring coordinates $\boldsymbol{c}$
Output: $\widehat{\mathbf{H}}_a$

```
Initialize parameters θ_m and θ_q;
for each epoch do
    Achieve HPC by concatenating H̃_t^↑ and P̃_t;
    Obtain latent code L by the encoding network;
    Calculate refined latent code N by latent transformer;
    for each pixel in coordinate c do
        Calculate Q_s from L;
        Calculate K_s and V_s from N and HPC;
        Compute V_c from L;
        Compute Q_c and K_c from N and HPC;
        Achieve ensemble features by internal and
        external transformer-ensembles;
        Obtain H̃ by the coordinate query mapping;
    end
    Calculate loss function L;
    Update parameters θ_m and θ_q;
end
Obtain model parameters θ_e^opt and θ_d^opt;
return Ĥ_a = Φ(c; r = Φ_m(H^↑, P; θ_m^opt), H̃^↑, P̃, θ_q^opt)
```

IV. Postintegrated ISSF

In this section, we design another ISSF called postISSF, in addition to preISSF developed in Section III. Fig. 6 graphically illustrates our postISSF, where the integration of spectral–spatial information is accomplished after ensemble feature extraction, while in the duration of coordinate query mapping. As shown in the figure, postISSF comprises a SpeTE, a SpaTE, and a spectral–spatial query fusion. We use the SpaTE and the SpeTE to extract spatial ensemble features and spectral ensemble features from both the latent code and the original input data, respectively. Then, we construct a spectral–spatial query fusion to integrate coordinate queries from spatial ensemble features and spectral ensemble features, so as to fulfill the final coordinate query mapping.

A. Spatial TE

SpaTE seeks to generate effective spatially context features from the latent code and the original PAN data, with both nonlocal similarities of pixels and the related high-resolution positional biases being utilized. To better extract such spatial ensemble features, PAN data, without HS data, are solely used as one of the sources of SpaTE since the former contains significantly more spatial information than the latter.

The structure of SpaTE can be found in Fig. 6. The latent code goes through a nonlocal spatial attention block to yield nonlocal spatial features before the convolutional operations to produce the key K_s and the value V_s. The structure of the nonlocal spatial attention block is graphically given in Fig. 7(a), where it achieves nonlocal attention by the combination of local self-attention and shifted windows. The query Q_s is produced by convolution operations connected to the latent code. Those convolution operations for Q_s, K_s, and V_s are similar to the spatial transformer-based upsampling we developed in pervious Section III (as shown in Fig. 3). Then, calculate the spatial weight map through the dot product of Q_s and K_s to measure the similarity between a Q_s pixel and its surrounding pixels K_s. Furthermore, we also add the implicit positional encoding here to utilize the positional bias to enhance the spatial weight map to obtain the final weight map. By multiplying the final weight map with V_s, preliminary spatial ensemble features are obtained. The preliminary spatial ensemble feature extraction can be mathematically represented as

$$\begin{aligned} Q_s &= f_{Qs}(L), \quad K_s^{(i,j)} = f_{Ks}(\mathcal{S}(L)), \quad V_s^{(i,j)} = f_{Vs}(\mathcal{S}(L)) \\ \delta_s &= \mathcal{P}\left(x_q - x^{(i,j)}\right) \\ E_{s1} &= \left(\mathrm{Softmax}\left(\frac{Q_s K_s^{(i,j)}}{\sqrt{C}}\right) \times \delta_s\right) \times V_s^{(i,j)} \end{aligned} \tag{20}$$

where C is the number of channels of the embedding Q_s, $\mathcal{P}(x) = [\sin(2^0 x), \cos(2^0 x), \ldots, \sin(2^{l-1} x), \cos(2^{l-1} x)]$ is the coordinate function, f_{Qs}, f_{Ks}, and f_{Vs} are convolution layers to yield Q_s, K_s, and V_s, $\mathcal{S}$ represents the nonlocal spatial attention block, and E_{s1} denotes preliminary spatial ensemble features.

Then, the spatial information in PAN data is also utilized in an attention manner. PAN data is directly used to generated the key K_s' and the value V_s', while the query Q_s' is formed from the latent code. By the dot production of Q_s' and K_s', we can achieve the similarity between the latent code and the original PAN data. Then, the implicit positional encoding involving the coordinate function $\mathcal{P}(x)$ is also used here to improve the spatial weight map. Finally, the preliminary spatial ensemble features from the PAN data are obtained by the multiplication of the final weight map and V_s'. The process can be formulated as

$$\begin{aligned} Q_s' &= f_{Qs}'(L), \quad K_s'^{(i,j)} = f_{Ks}(\mathbf{P}), \quad V_s'^{(i,j)} = f_{Vs}'(\mathbf{P}) \\ \delta &= \mathcal{P}\left(x_q - x^{(i,j)}\right) \\ E_{s2} &= \left(\mathrm{Softmax}\left(\frac{Q_s' K_s'^{(i,j)}}{\sqrt{C}}\right) \times \delta\right) \times V_s'^{(i,j)} \end{aligned} \tag{21}$$

where C is the channel number of Q_s', $\mathcal{P}(x)$ is the coordinate function, and f_{Qs}', f_{Ks}', and f_{Vs}' are convolution layers to

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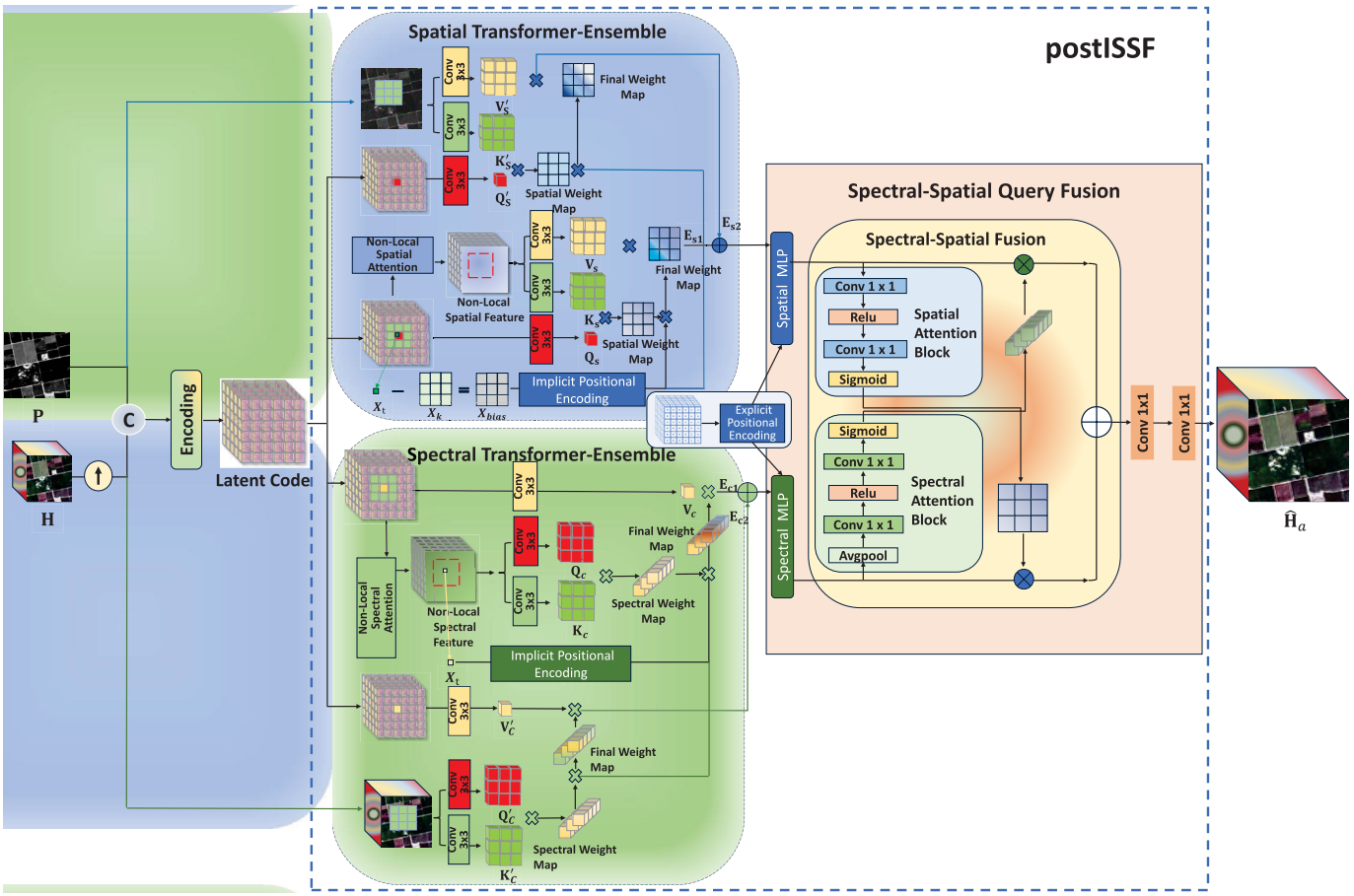


Fig. 6. Overall architecture of postISSF. SpeTE module extracts spectral ensemble features of nonlocal spectral context from the latent code and the original HS data, SpaTE generates spatial ensemble features of spatial context from the latent code and the original PAN data, and spectral–spatial query fusion integrates both spatial and spectral ensemble features to fulfill coordinate query mapping.

produce Q'_s, K'_s, and V'_s, and E_{s2} represents the preliminary spatial ensemble features associated with PAN.

The final spatial ensemble features E_s are synthesized by the preliminary spatial ensemble features E_{s1} and E_{s2} as follows:

$$E_s = E_{s1} + E_{s2}. \tag{22}$$

B. Spectral TE

Complementary to SpaTE which refines spatial context, here we design a SpeTE module which extracts the nonlocal spectral context information from the latent code and the original HS data. To effectively extract the spectral ensemble features, HS data are directly used as one of the sources of SpeTE, as the data contain abundant spectral information.

The structure of the SpeTE module can be found in Fig. 6. Basically, three convolution layers are used to calculate Q_c, K_c, and V_c. V_c is generated from the original latent code, where Q_c and K_c are produced from the nonlocal spectral feature, which is spectrally modulated by a nonlocal spectral attention block. Different from the SpaTE module, here V_c represents pixels to be restored, each of which is the pixel x_v in Fig. 4, while Q_c and K_c denote the pixels in their surrounding region. In SpeTE, the multiplication of Q_c and K_c is conducted along the spectral dimension, calculating the spectral correlation within a pixel's surrounding region. Then, the implicit positional encoding is also used here to strengthen the spectral weight map. The final weight map obtained is subsequently multiplied with V_c to tune the spectral information of each pixel. Thus, the process to yield preliminary spectral

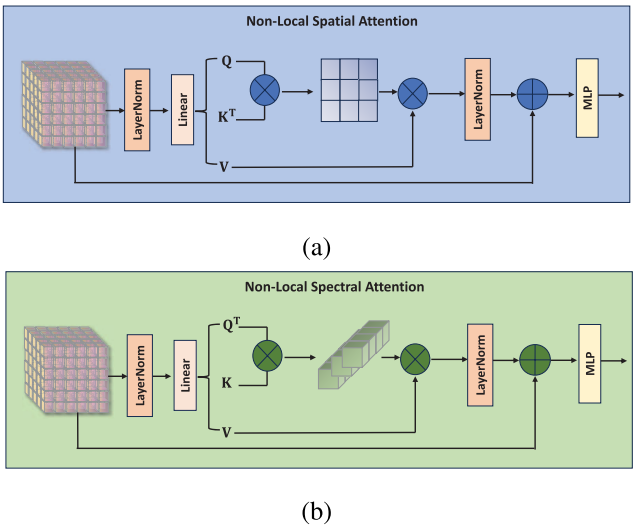


Fig. 7. Structures of nonlocal spatial/spectral attention blocks. (a) Spatial attention and (b) spectral attention.

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ensemble features from the latent code can be formulated as

$$V_c = f_{Vc}(L), \quad K_c^{(i,j)} = f_{Kc}(\mathcal{C}(L)), \quad Q_c^{(i,j)} = f_{Qc}(\mathcal{C}(L))$$
$$\delta = \mathcal{P}(x_q)$$
$$E_{c1} = \left(\text{Softmax}\left(\frac{Q_c^{(i,j)} K_c^{(i,j)}}{\sqrt{C}}\right) \times \delta\right) \times V_c \tag{23}$$

where $\mathcal{P}(x)$ is the coordinate function, f_{Qc}, f_{Kc}, and f_{Vc} are convolution layers to yield Q_c, K_c, and V_c, $\mathcal{C}$ represents the nonlocal spectral attention operation, and E_{c1} denotes the preliminary spectral ensemble features.

In addition, we also utilize the spectral information of the original HS data to improve SpeTE. HS data are first used to yield Q_c' and K_c', and the spectral weight map is calculated by the dot production of Q_c' and K_c'. Then, the implicit positional encoding is employed to strengthen the spectral weight map. Finally, V_c' produced by the latent code is multiplied by the final weight map to obtain the preliminary spectral ensemble features from the HS data. The process can be formulated as follows:

$$V_c' = f_{Vc}'(L), \quad K_c'^{(i,j)} = f_{Kc}'(\mathbf{H}), \quad Q_c'^{(i,j)} = f_{Qc}'(\mathbf{H})$$
$$\delta = \mathcal{P}(x_q)$$
$$E_{c2} = \left(\text{Softmax}\left(\frac{Q_c'^{(i,j)} K_c'^{(i,j)}}{\sqrt{C}}\right) \times \delta\right) \times V_c' \tag{24}$$

where $\mathcal{P}(x)$ is the coordinate function, f_{Qc}, f_{Kc}, and f_{Vc} are convolution layers, and E_{c2} represents the preliminary spectral ensemble features.

The final spectral ensemble features E_c are formed with the preliminary spectral ensemble features E_{c1} and E_{c2} as follows:

$$E_c = E_{c1} + E_{c2}. \tag{25}$$

C. Spectral–Spatial Coordinate Query Fusion

As previously designed, we have a SpaTE and a SpeTE to achieve effective spatial ensemble features and spectral ensemble features, respectively. Then, we need to develop a process to integrate those features to fulfill coordinate query mapping to produce the desired high-resolution HS data. Here, we design a spectral–spatial coordinate query fusion method, where a preliminary spectral query mapping and a preliminary spatial query mapping are performed first and then they are processed with a query mapping-level fusion to accomplish a final spectral–spatial query mapping.

1) Spatial Query Mapping: Spatial query mapping seeks to decode the spatial ensemble features to build a representation of HS data in a spatial continuous manner. Similar to the coordinate query mapping in preISSF, explicit positional encoding is used to produce a coordinate manifold, which is combined with spatial ensemble features to be input to a spatial MLP. The spatial MLP then generates a preliminary continuous HS estimation $\mathbf{O}_s$. The process can be formulated as

$$\mathbf{O}_s = \mathcal{M}_s(E_s, \boldsymbol{c}_{\text{pe}}) \tag{26}$$

where $\mathcal{M}_s$ stands for the spatial MLP and $\boldsymbol{c}_{\text{pe}}$ represents the explicit coordinate manifold built with the explicit coordinate function $\mathcal{P}'(x)$.

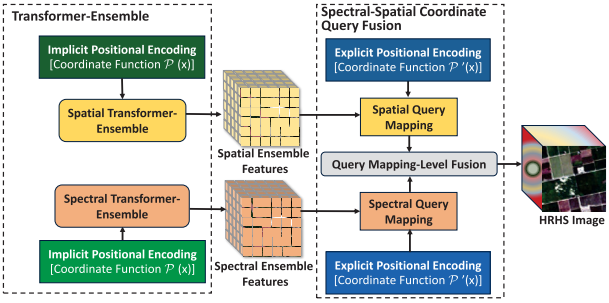


Fig. 8. Explicit–implicit positional encoding system of postISSF.

2) Spectral Query Mapping: In contrast to spatial query mapping, the spectral query mapping is built to use the SpeTE features to represent HS data in a spectral continuous manner. Similarly, the explicit positional encoding is also employed to generate an explicit coordinate manifold $\boldsymbol{c}_{\text{pe}}$ for a spectral MLP, and then the spectral MLP generates a preliminary continuous HS estimation $\mathbf{O}_c$. The process can be formulated as

$$\mathbf{O}_c = \mathcal{M}_c(E_c, \boldsymbol{c}_{\text{pe}}) \tag{27}$$

where $\mathcal{M}_c$ stands for the spectral MLP.

3) Query Mapping-Level Fusion: The above spatial query mapping focuses on the spatial reconstruction, while the spectral query mapping aims at restoring the spectral information. Thus, to obtain the final spectral–spatial coordinate query mapping, we use a query mapping-level fusion to fuse the preliminary spatial query mapping and the preliminary spectral query mapping. As can be found in Fig. 6, the output of the spatial query mapping $\mathbf{O}_s$ goes through a spatial attention block to generate the spatial attention map AM_s, while the output of the spectral query mapping $\mathbf{O}_c$ is fed into a spectral attention block to yield the spectral attention map AM_c. Then, $\mathbf{O}_s$ is spectral attention modulated by AM_c, while AM_s is used to spatially enhanced $\mathbf{O}_c$. Finally, they are added up and go through two 1×1 convolution layers to reconstruct the final target image $\widehat{\mathbf{H}}_a$ with the desired resolution. The fusion process can be mathematically represented as

$$\text{AM}_s = \text{Sigmoid}(\mathcal{F}(\text{Relu}(\mathcal{F}(\mathbf{O}_s))))$$
$$\text{AM}_c = \text{Sigmoid}(\mathcal{F}(\text{Relu}(\mathcal{F}(\text{Avg pool}(\mathbf{O}_c)))))$$
$$\widehat{\mathbf{H}}_a = \mathcal{F}(\mathcal{F}(\text{AM}_s \times \mathbf{O}_c + \text{AM}_c \times \mathbf{O}_s)) \tag{28}$$

where $\mathcal{F}$ denotes the convolution layer with the kernel size of 1×1.

It is noteworthy that the above spatial query mapping and spectral query mapping in the coordinate query fusion comprise explicit positional encoding processes, which combine with the implicit positional encoding in proceeding TE to compose the explicit–implicit positional encoding system of postISSF (as shown in Fig. 8). Overall, the implementation of postISSF can be summarized as the following Algorithm 2.

V. Experimental Results

In this section, we will introduce the experimental results to validate the effectiveness of our proposed method.

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TABLE I
PROPERTIES OF THE DATASETS

Datasets	Data size	Spectral range	Bands channel	Instrument
Pavia Center	HS 1000×625	0.4-0.8 μm	102	ROSIS
Merced	HS 2100×180, PAN 6300×540	0.4-2.5 μm	134	Hyperion and ALI
Changping	HS 2900×180, PAN 8700×540	0.4-2.5 μm	137	Hyperion and ALI

Algorithm 2 postISSF

Input: $\widetilde{\mathbf{H}}_t^{\uparrow}$, $\widetilde{\mathbf{P}}_t$, $\mathbf{H}_t$, $\mathbf{H}^{\uparrow}$, $\mathbf{P}$, coordinate $\boldsymbol{c}$
Output: $\widehat{\mathbf{H}}_a$

Initialize parameters $\boldsymbol{\theta}_m$ and $\boldsymbol{\theta}_q$;
for *each epoch* **do**
  Obtain latent code L by the encoding network;
  for *each pixel in coordinate* $\boldsymbol{c}$ **do**
    Obtain non-local spatial feature S by non-local spatial attention;
    Obtain non-local spectral feature C by non-local spectral attention;
    Calculate Q_s from L;
    Calculate K_s and V_s from S and $\widetilde{\mathbf{P}}_t$;
    Compute V_c from L; Compute Q_c and K_c from C and $\mathbf{H}_t$;
    Achieve spatial transformer-ensemble features by Q_s, K_s and V_s;
    Achieve spectral transformer-ensemble features by Q_c, K_c and V_c;
    Obtain $\widetilde{\mathbf{H}}$ by spectral-spatial coordinate query fusion;
  end
  Calculate loss function $\mathcal{L}$;
  Update parameters $\boldsymbol{\theta}_m$ and $\boldsymbol{\theta}_q$;
end
Obtain model parameters $\boldsymbol{\theta}_m^{opt}$ and $\boldsymbol{\theta}_q^{opt}$;
return $\widehat{\mathbf{H}}_a = \Phi(\boldsymbol{c}; \boldsymbol{r} = \Phi_m(\mathbf{H}^{\uparrow}, \mathbf{P}; \boldsymbol{\theta}_m^{opt}), \widetilde{\mathbf{H}}^{\uparrow}, \widetilde{\mathbf{P}}, \boldsymbol{\theta}_q^{opt})$

A. Experimental Datasets

We test our proposed ISSFs on three datasets, including the Changping dataset, the Merced dataset, and the Pavia Center Dataset [30], [46]. The Changping and the Merced are both real datasets, while the Pavia Center is a simulated dataset. Some descriptions of the datasets are listed in Table I.

1) Pavia Center Dataset: The orignial size of the dataset is 1000×625, while the top part of $[0, 1000] \times [0, 425]$ is selected for training, the bottom left part of $[0, 800] \times [425, 625]$ is used for validation and the rest part of $[800, 1000] \times [425, 625]$ is for test. Totally, 7680 training samples and 768 validation samples with sizes of 20×20 are generated by the involving of data augmentation (including rotation and flipping).

2) Merced Dataset: The orignial size of the dataset is 2100×180, while the part of $[1300, 1900] \times [0, 180]$ is selected for training, the part of $[900, 1300] \times [0, 180]$ is used for validation and the part of $[1920, 2100] \times [0, 180]$ is for test. Totally, 9600 training samples and 960 validation samples with sizes of 20×20 are produced.

3) Changping Dataset: The orignial size of the dataset is 2900×180, while the part of $[1700, 2900] \times [0, 180]$ is selected for training, the part of $[1100, 1700] \times [0, 180]$ is used for validation and the part of $[920, 1100] \times [0, 180]$ is for test. 9600 training samples and 960 validation samples with sizes of 20×20 are generated for our experiments.

B. Experiments Setting

Both standard HS pansharpening and ARHS pansharpening are conducted in our experiments to validate the effectiveness of our proposed methods. In those two categories of experiments, both reduced-resolution quantitative assessments [47] and full-resolution quantitative assessments [48], [49] are used to evaluate the methods. Reduced-resolution quantitative assessments includes cross correlation (CC), spectral angle mapper (SAM), root-mean-square error (RMSE), and the extended universal image quality index ($Q2^n$), which calculate the assessments between pansharpening results and label images. Full resolution quantitative assessments are used to evaluate the pansharpening results of real PAN data and real HS data, containing spatial distortion index D_s, spectral distortion index D_λ, the quality with on reference (QNR) and hybrid QNR (HQNR).

Some existing HS pansharpening methods are compared with our proposed methods, including six traditional methods, four deep neural network methods, and two ARHS pansharpening methods. Traditional methods comprise Bicubic interpolation, SFIM, GSA, MTF-GLP, BayesSparse, and GFPCA [5], [50], [51], [52], while the deep neural network methods contain PNN [17], HyperPNN [22], HSpeNet [23], and DSNet [21]. These ten methods are standard pansharpening methods, which cannot be directly used for the ARHS pansharpening task. Therefore, in ARHS pansharpening experiments, bicubic interpolation is incorporated into these techniques as a final pansharpening step, enabling them to achieve ARHS pansharpening, which are called X-Bic, like PNN-Bic. Two ARHS pansharpening methods concludes ARHSCNN ARHS-CNN [29] and VSPC-DARN, where DARN [24] is used as the encoding network and VSPC [30] is the decoding network in VSPC-DARN. It is noteworthy that in the experiments we use three encoding networks, including DARN, RHDNet [25], and IC-DyPNN [6], as preliminary networks for our preISSF and postISSF. Therefore, six pansharpening networks developed by us are in the experimental comparison, which we call preISSF1, preISSF2, preISSF3, postISSF1, postISSF2, and postISSF3, respectively. We conducted all the neural network experiments using an

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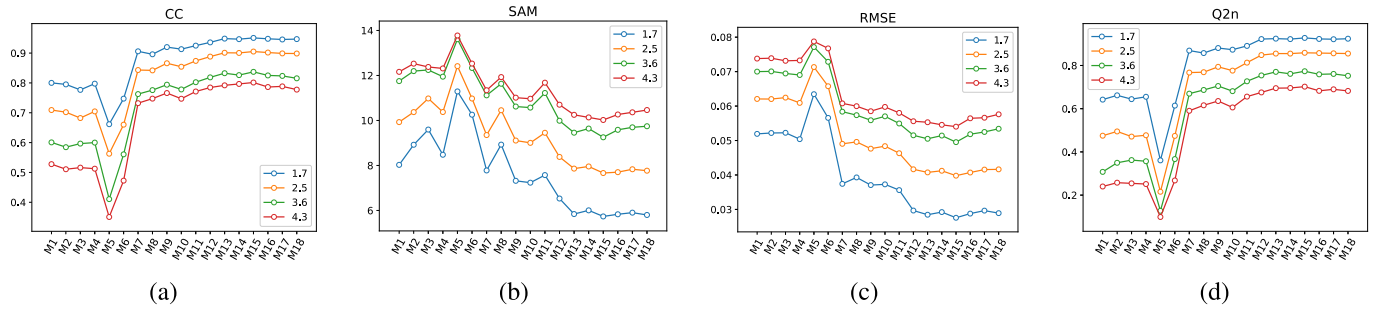


Fig. 9. Pansharpening performances under arbitrary scales s_a on PaviaC dataset. (a) CC, (b) SAM, (c) RMSE, and (d) Q2n. M1–M18 represent Bicubic, GSA-Bic, SFIM-Bic, MTF-GLP-Bic, GFPCA-Bic, BayesSparse-Bic, PNN-Bic, HyperPNN-Bic, HspeNet-Bic, DSNet-Bic, ARHSCNN, VSPC-DARN, preISSF1, preISSF2, preISSF3, postISSF1, postISSF2, and postISSF3, respectively.

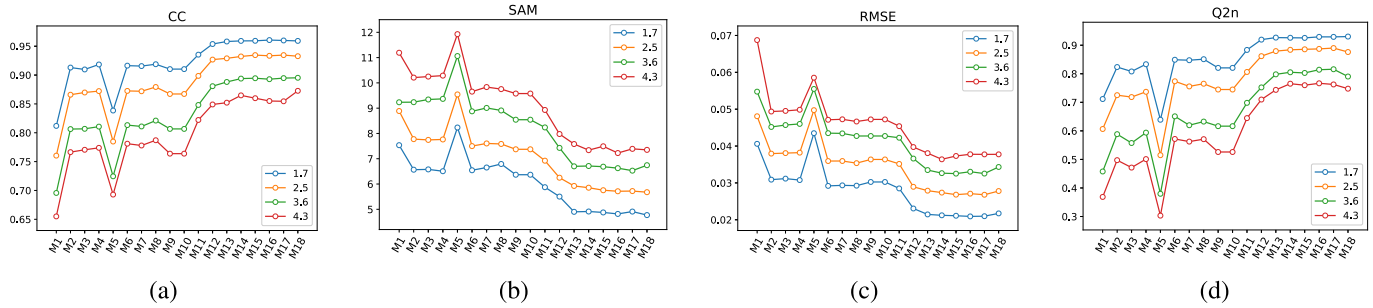


Fig. 10. Pansharpening performances under arbitrary scales s_a on Merced dataset. (a) CC, (b) SAM, (c) RMSE, and (d) Q2n.

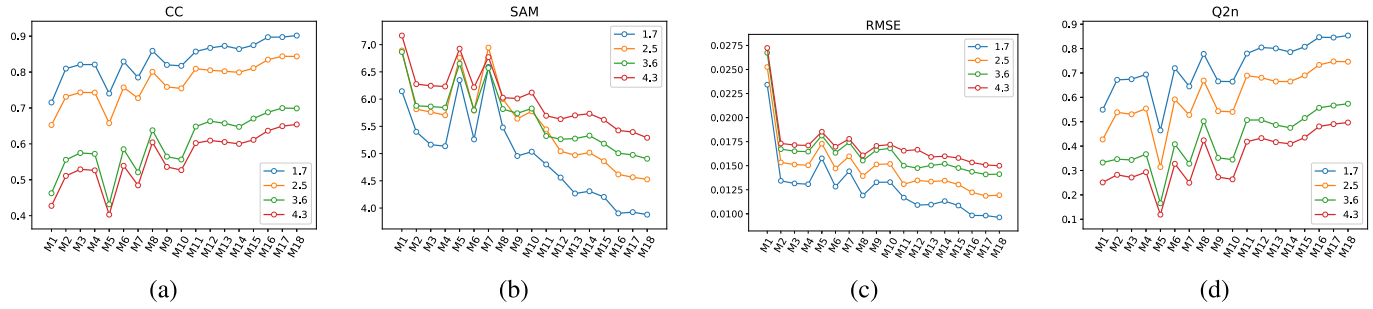


Fig. 11. Pansharpening performances under arbitrary scales s_a on Changping dataset. (a) CC, (b) SAM, (c) RMSE, and (d) Q2n.

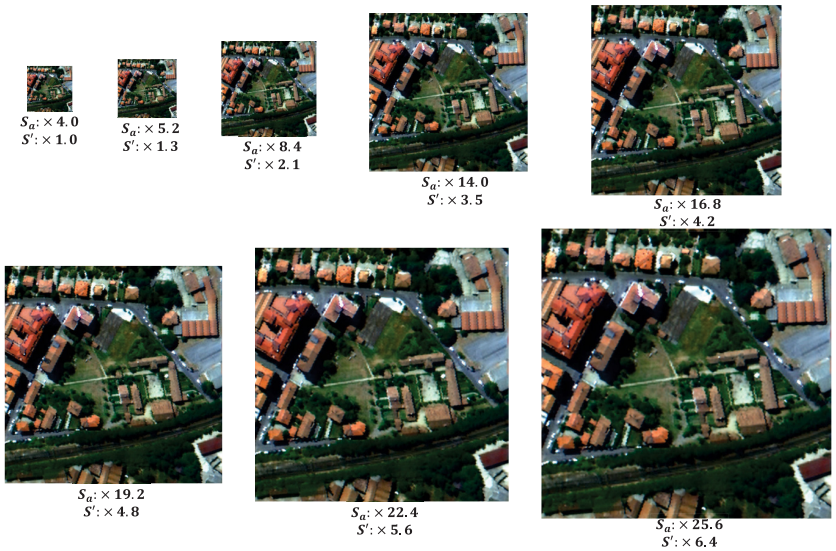


Fig. 12. Visualization of ARHS pansharpening by using postISSF3 on the reduced-resolution PaviaC dataset. The excellent visual effect of ARHS pansharpening for PaviaC is confirmed.

NVIDIA GeForce RTX 3060 SUPER GPU. The activation function used in the experiments is the RELU Function, and the loss function used in the experiments is $l1$ function. The total training epoch is 300 with a batch size of 64.

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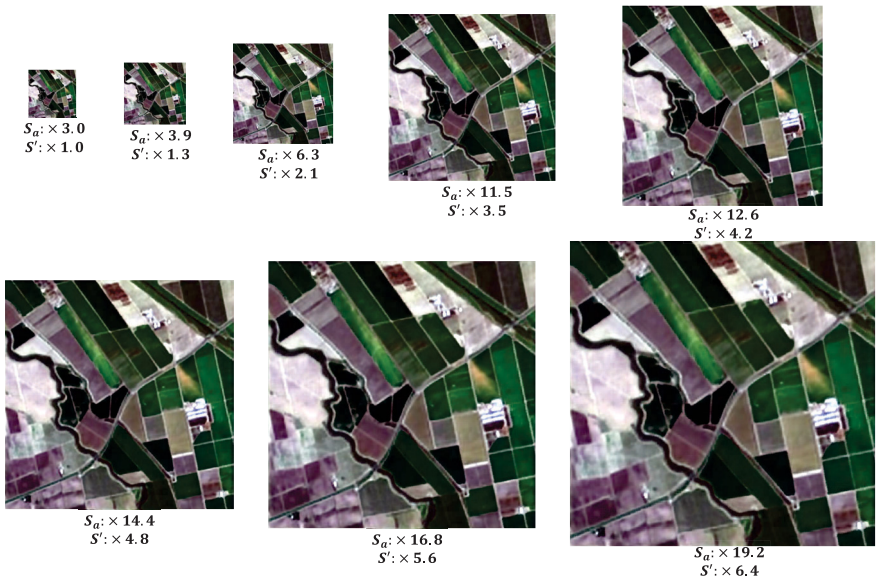


Fig. 13. Visualization of ARHS pansharpening by using postISSF3 on the reduced-resolution Merced dataset. The excellent visual effect of ARHS pansharpening for Merced is confirmed.

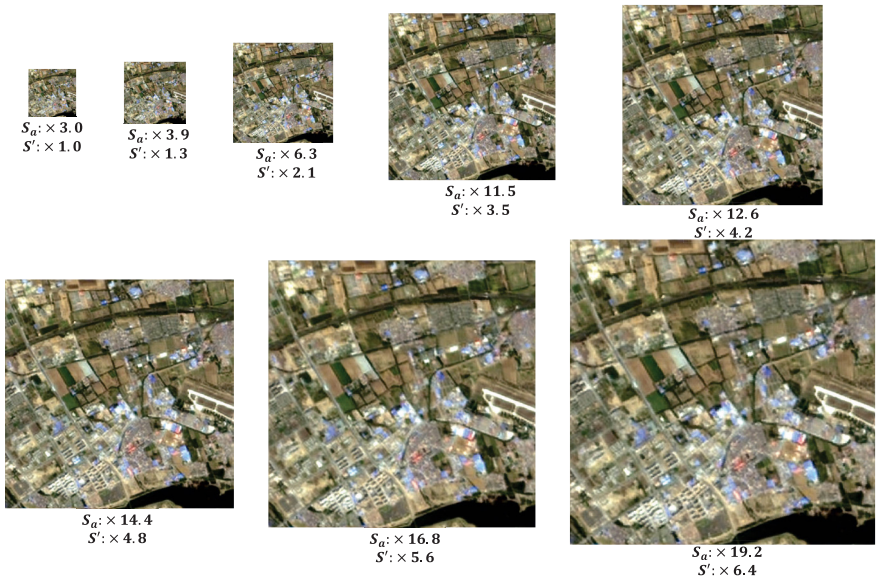


Fig. 14. Visualization of ARHS pansharpening by using postISSF3 on the reduced-resolution Changping dataset. The excellent visual effect of ARHS pansharpening for Changping is confirmed.

C. ARHS Pansharpening Experiments

ARHS pansharpening experiments are used to evaluate the pansharpening results with any spatial resolution, which include reduced-resolution experiments and full-resolution experiments.

1) Reduced-Resolution Experiments: We test ten compared methods and our proposed methods on the reduced-resolution experiments first. In the experiments, we set $s' = 1.7, 2.5, 3.6,$ and 4.3 for intermediate scales to make quantitative assessment comparison and visualization on three datasets. In the reduced-resolution experiments, we used the observed LRHS images as the ground-truth images, the input data are generated by downsampling the observed LRHS and the observed PAN. Then, the results of these methods are able to be assessed by quantitative assessments since the ground-truth images are available. As shown in Figs. 9–11, both preISSF methods and postISSF methods perform much better than other pansharpening methods with all scales on three datasets. In addition, we find out that the ARHS pansharpening methods outperforms those bicubic upsampling-based pansharpening methods, especially with the larger scale s'.

Then, we perform a visual evaluation on the PaviaC dataset and the Merced dataset. Figs. 12–14 show the visual results of preISSF and postISSF, where we set $s' = 1.0, 1.3, 2.1, 3.5, 4.2, 4.8, 5.6,$ and 6.4. The pansharpening results of our preISSF methods and postISSF methods keep high spectral–spatial fidelity even if the scale becomes larger.

2) Full-Resolution Experiments: Full-resolution experiments are used to verify the effectiveness of our

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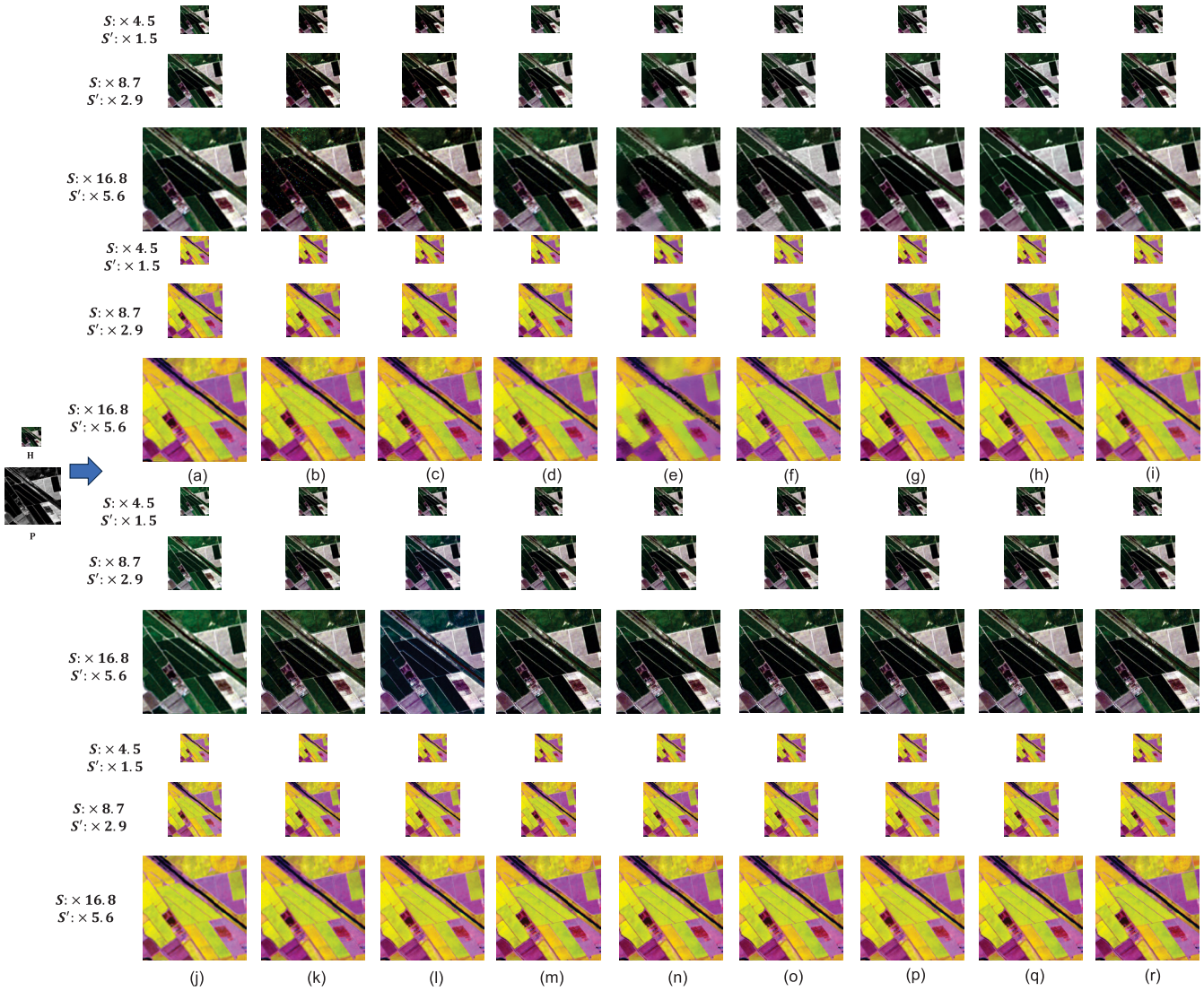


Fig. 15. Visualization of ARHS pansharpening on the real Merced dataset. (a) Bicubic, (b) SFIM-Bic, (c) MTF-GLP-Bic, (d) GSA-Bic, (e) GFPCA-Bic, (f) BayesSparse-Bic, (g) PNN-Bic, (h) HyperPNN-Bic, (i) HSpeNet-Bic, (j) DSNet-Bic, (k) ARHSCNN, (l) VSPC-DARN, (m) preISSF1, (n) preISSF2, (o) preISSF3, (p) postISSF1, (q) postISSF2, and (r) postISSF3.

proposed methods when applying ARHS pansharpening on the observed LRHS data and the observed PAN data. Similar to the reduced-resolution experiments, all pansharpening methods are compared on two datasets, i.e., the Merced dataset and the Changping dataset. In the experiments, we set $s' = 1.5, 2.9$, and 5.6 for intermediate scales to make a visual comparison. Figs. 15 and 16 give RGB images composed of three visible bands and false-color images synthesized by three bands out of the visible range. For more clear graphical illustration, Fig. 17 displays the enlarged views of a local area (marked with red rectangles in Fig. 16). Through close inspection on these figures, especially on Fig. 17, we could observe that our perISSF and postISSF methods achieve better arbitrary resolution results. Bicubic, BayesSparse, and DSNet methods yield obvious vague results, which implies dropping many spatial details during the HRHS reconstruction. BayesSparse and HyperPNN methods incur obvious spectral distortion. Methods specifically designed for ARHS pansharpening, including ARHS-CNN, VSPC-DARN, preISSF3, and postISSF3, achieve relatively better ARHS pansharpening results, while VSPC-DARN still produces more spectral distortion, and ARHS-CNN yields more blurring details than preISSF3 and postISSF3. From the experimental results, the excellent ARHS pansharpening performances of our newly designed ARHS pansharpening methods are verified.

D. Standard HS Pansharpening Experiments

We additionally conduct standard HS pansharpening experiments to validate that our methods still exhibit good performance for the standard HS pansharpening task. The experiments also include reduced-resolution experiments and full-resolution experiments.

1) Reduced-Resolution Experiments: The experimental results are with the same spatial resolution of the observed

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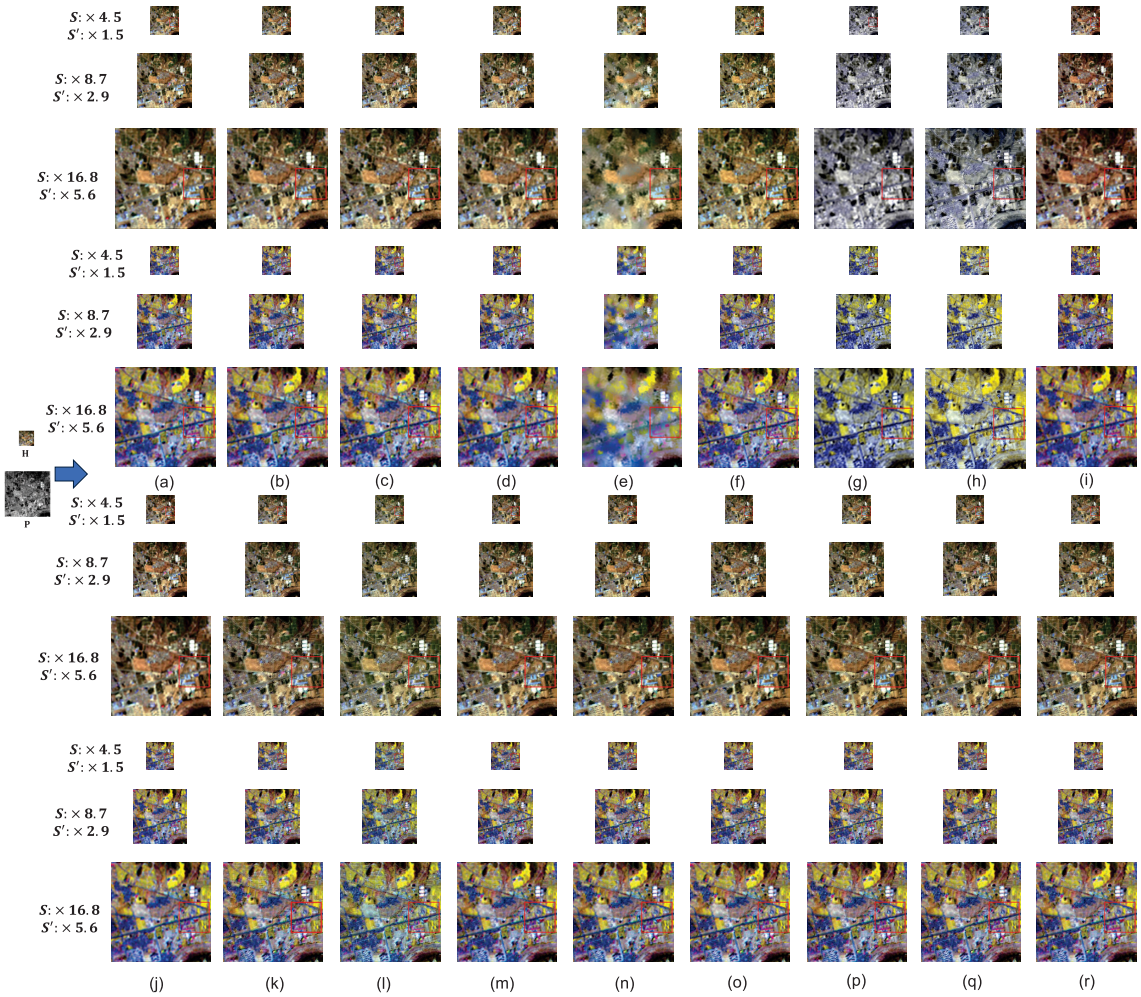


Fig. 16. Visualization of ARHS pansharpening on the real Changping dataset. (a) Bicubic, (b) SFIM-Bic, (c) MTF-GLP-Bic, (d) GSA-Bic, (e) GFPCA-Bic, (f) BayesSparse-Bic, (g) PNN-Bic, (h) HyperPNN-Bic, (i) HSpeNet-Bic, (j) DSNet-Bic, (k) ARHSCNN, (l) VSPC-DARN, (m) preISSF1, (n) preISSF2, (o) preISSF3, (p) postISSF1, (q) postISSF2, and (r) postISSF3.

TABLE II
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE PAVIAC DATASET

Methods	CC(↑)	SAM(↓)	RMSE(↓)	Q2n(↑)
Bicubic	0.8269	7.1396	0.0434	0.7786
GSA	0.8755	7.0317	0.0406	0.8041
MTF-GLP	0.8811	6.2433	0.0380	0.8114
SFIM	0.8589	7.7595	0.0407	0.7915
GFPCA	0.7699	9.3132	0.0513	0.5621
BayesSparse	0.8292	8.3741	0.0465	0.7366
PNN	0.9527	6.3211	0.0257	0.9347
HyperPNN	0.9389	7.3003	0.0288	0.9218
HSpeNet	0.9603	5.5275	0.0257	0.9423
DSNet	0.9593	5.4458	0.0250	0.9417
ARHSCNN	0.9619	5.6992	0.0245	0.9416
VSPC-DARN	0.9699	5.0827	0.0204	0.9588
preISSF1	0.9728	4.5589	0.0192	0.9613
preISSF2	0.9732	4.6011	0.0194	0.9617
preISSF3	0.9748	4.4818	0.0187	0.9639
postISSF1	<u>0.9753</u>	<u>4.4161</u>	<u>0.0185</u>	<u>0.9643</u>
postISSF2	0.9739	4.4723	0.0192	0.9629
postISSF3	**0.9759**	**4.3323**	**0.0182**	**0.9658**

TABLE III
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE MERCED DATASET

Methods	CC(↑)	SAM(↓)	RMSE(↓)	Q2n(↑)
Bicubic	0.9006	4.4884	0.0306	0.8789
GSA	0.9607	4.9418	0.0210	0.9157
MTF-GLP	0.9635	4.8459	0.0210	0.9229
SFIM	0.9441	5.1678	0.0220	0.8939
GFPCA	0.9019	5.6779	0.0346	0.7778
BayesSparse	0.9571	5.5110	0.0205	0.9188
PNN	0.9586	5.4438	0.0202	0.9209
HyperPNN	0.9587	5.7727	0.0210	0.9199
HSpeNet	0.9688	4.4582	0.0193	0.9408
DSNet	0.9632	4.4464	0.0192	0.9221
ARHSCNN	0.9684	4.7169	0.0200	0.9371
VSPC-DARN	0.9702	5.3024	0.0183	0.9451
preISSF1	0.9761	3.9951	0.0159	0.9548
preISSF2	0.9772	3.9653	0.0155	0.9540
preISSF3	<u>0.9794</u>	<u>3.8028</u>	<u>0.0148</u>	<u>0.9592</u>
postISSF1	0.9789	3.8123	0.0149	0.9578
postISSF2	0.9780	3.9222	0.0152	0.9564
postISSF3	**0.9808**	**3.6516**	**0.0140**	**0.9627**

LRHS images. Thus, we merely set $s' = 1$ for ARHS pansharpening methods to achieve standard pansharpening. We conduct the experiments on three datasets, i.e., the Merced dataset, the Changping dataset, and the Pavia Center

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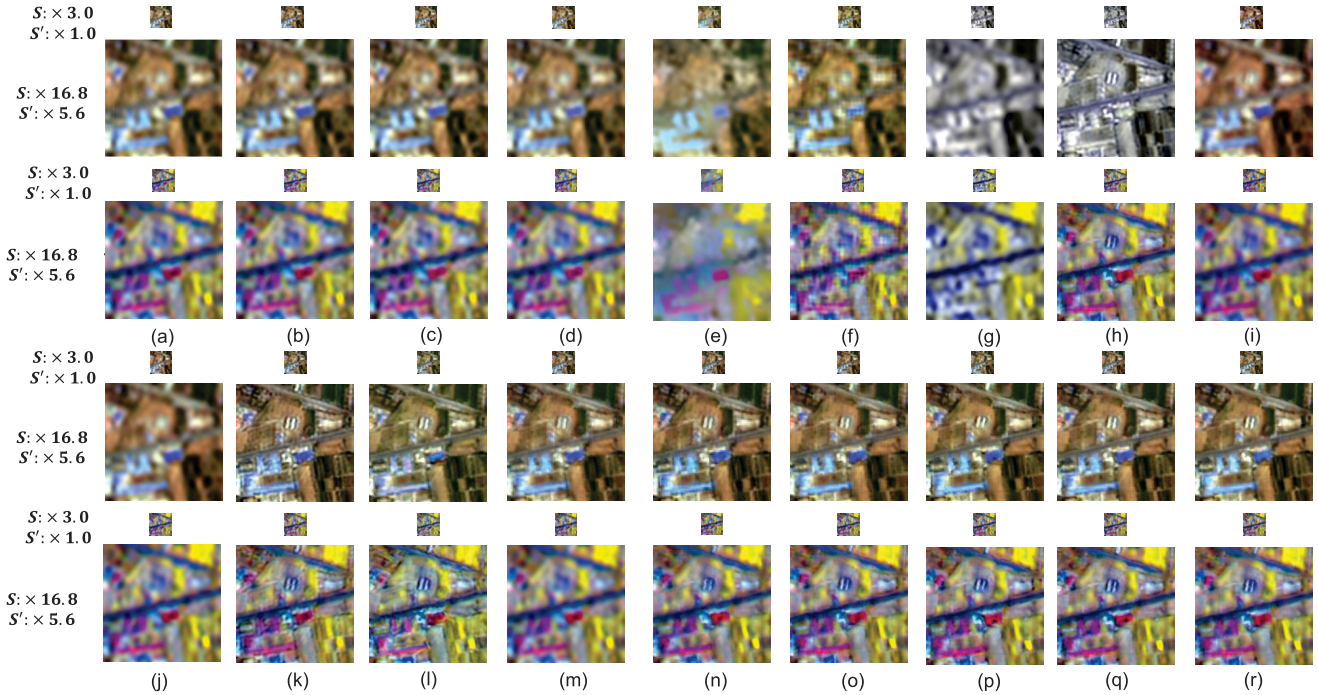


Fig. 17. Enlarged rectangle area of Changping dataset. (a) Bicubic, (b) SFIM-Bic, (c) MTF-GLP-Bic, (d) GSA-Bic, (e) GFPCA-Bic, (f) BayesSparse-Bic, (g) PNN-Bic, (h) HyperPNN-Bic, (i) HSpeNet-Bic, (j) DSNet-Bic, (k) ARHSCNN, (l) VSPC-DARN, (m) preISSF1, (n) preISSF2, (o) preISSF3, (p) postISSF1, (q) postISSF2, and (r) postISSF3.

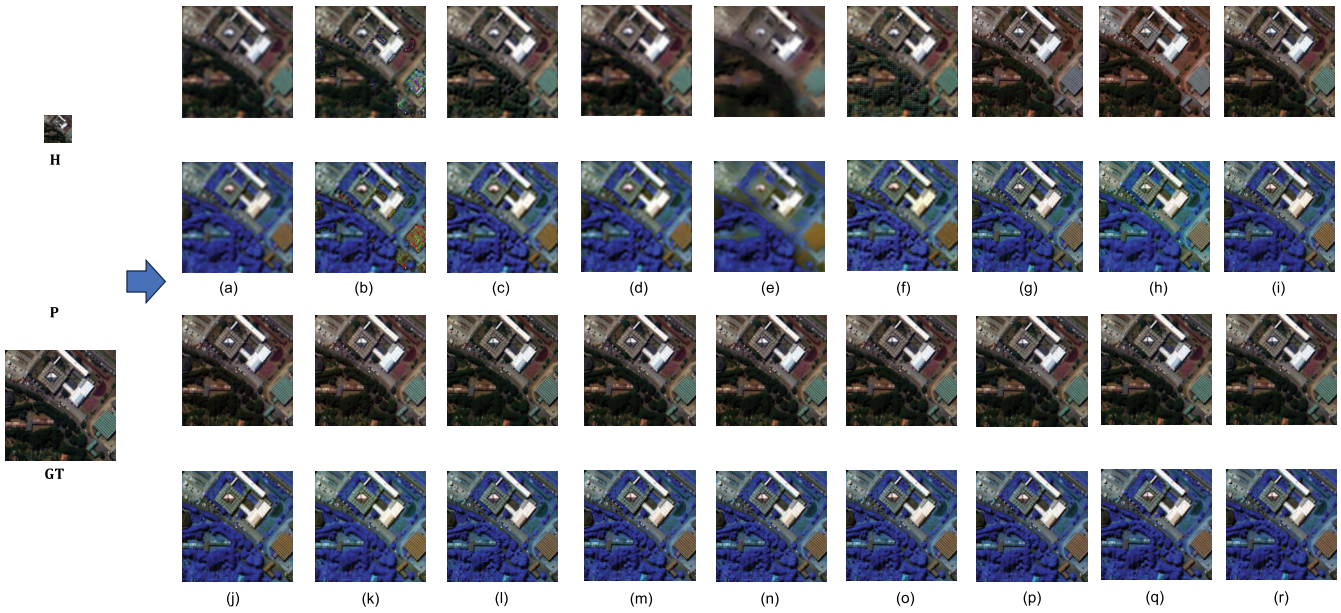


Fig. 18. Visualization of standard HS pansharpening on the simulated PaviaC dataset. (a) Bicubic, (b) SFIM, (c) MTF-GLP, (d) GSA, (e) GFPCA, (f) BayesSparse, (g) PNN, (h) HyperPNN, (i) HSpeNet, (j) DSNet, (k) ARHSCNN, (l) VSPC-DARN, (m) preISSF1, (n) preISSF2, (o) preISSF3, (p) postISSF1, (q) postISSF2, and (r) postISSF3.

dataset. Similarly, four quantitative assessments, i.e., CC, SAM, ERGAS, and $Q2^n$ are used to validate the effectiveness of our methods. As shown in Tables II–IV, postISSF methods perform best on all datasets, while preISSF methods achieve the second place in the quantity assessment. Then, from the Fig. 18, the RGB results of our methods are most similar to the ground-truth image, which validates the effectiveness of our methods.

2) Full-Resolution Experiments: The experiments are conducted on two datasets, i.e., the Changping dataset and the Merced dataset, which perform standard HS pansharpening on the observed HS data and the observed PAN data. Also,

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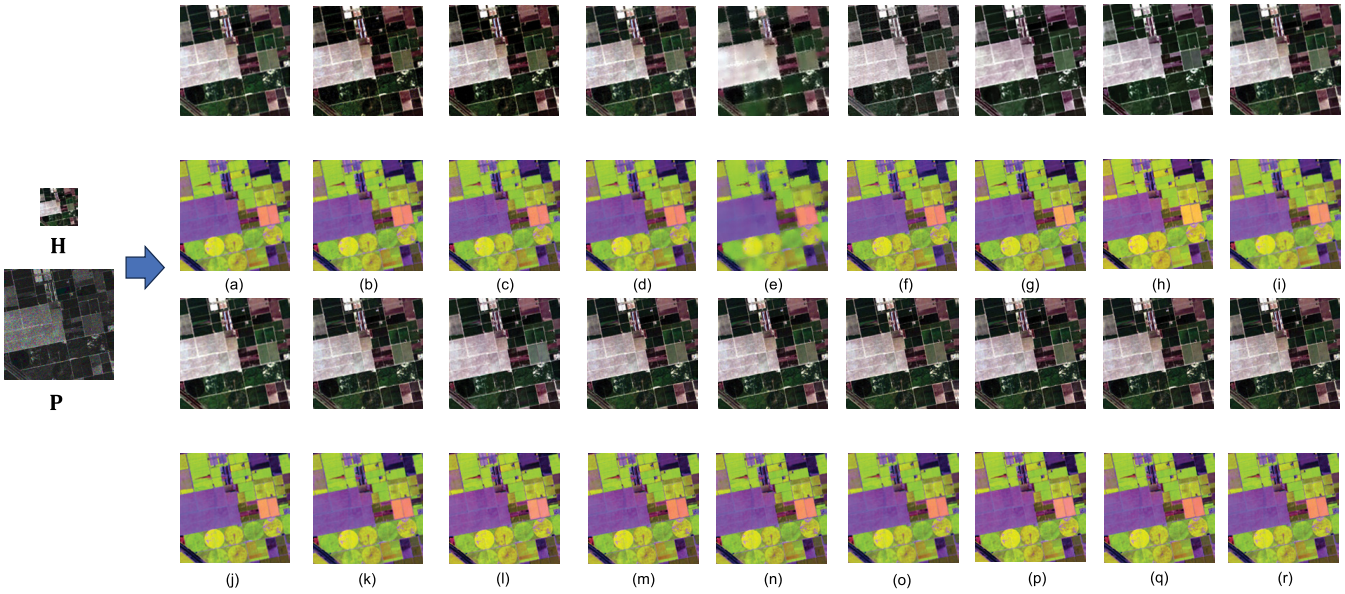


Fig. 19. Visualization of standard HS pansharpening on the real Merced dataset. (a) Bicubic, (b) SFIM, (c) MTF-GLP, (d) GSA, (e) GFPCA, (f) BayesSparse, (g) PNN, (h) HyperPNN, (i) HSpeNet, (j) DSNet, (k) ARHSCNN, (l) VSPC-DARN, (m) preISSF1, (n) preISSF2, (o) preISSF3, (p) postISSF1, (q) postISSF2, and (r) postISSF3.

TABLE IV
QUALITY METRICS OF STANDARD HS PANSHARPENING ON THE CHANGPING DATASET

Methods	CC(↑)	SAM(↓)	RMSE(↓)	Q2n(↑)
Bicubic	0.8810	5.1903	0.0129	0.7721
GSA	0.9060	4.4800	0.0101	0.8263
MTF-GLP	0.9152	4.1721	0.0096	0.8462
SFIM	0.9131	4.2270	0.0098	0.8319
GFPCA	0.8403	5.7502	0.0133	0.6281
BayesSparse	0.9194	4.3627	0.0094	0.8613
PNN	0.8799	6.5056	0.0117	0.7891
HyperPNN	0.9144	5.5009	0.0100	0.8609
HSpeNet	0.9118	4.2405	0.0101	0.8719
DSNet	0.9122	4.4188	0.0099	0.8747
ARHSCNN	0.9366	4.1763	0.0843	0.8932
VSPC-DARN	0.9308	4.1751	0.0088	0.8829
preISSF1	0.9548	3.2990	0.0070	0.9236
preISSF2	0.9559	3.2570	0.0069	0.9260
preISSF3	0.9557	3.2820	0.0070	0.9218
postISSF1	0.9590	3.1969	0.0068	0.9290
postISSF2	0.9581	3.2027	0.0067	0.9294
postISSF3	**0.9598**	**3.1319**	**0.0066**	**0.9314**

TABLE V
QUALITY METRICS OF FULL-RESOLUTION EXPERIMENT ON THE MERCED DATASET

Methods	$D_\lambda^F(\downarrow)$	$D_\lambda(\downarrow)$	$D_s(\downarrow)$	QNR(↑)	HQNR(↑)
Bicubic	0.1988	0.1874	0.1841	0.6630	0.6536
GSA	0.1791	0.1755	0.1717	0.6829	0.0.6799
SFIM	0.1581	0.1525	0.1701	0.7033	0.6987
MTF-GLP	0.1621	0.1577	0.1642	0.6975	0.7003
GFPCA	0.2241	0.2164	0.1705	0.6499	0.6436
BayesSparse	0.1701	0.1528	0.1738	0.6999	0.6857
PNN	0.0781	0.0728	0.0842	0.8491	0.8443
HyperPNN	0.0811	0.0729	0.0812	0.8518	0.8443
HSpeNet	0.0591	0.0401	0.0729	0.8899	0.8723
DSNet	0.0761	0.0699	0.0833	0.8526	0.8469
ARHSCNN	0.0537	0.0406	0.0746	0.8877	0.8757
VSPC-DARN	0.0713	0.0636	0.0893	0.8527	0.8458
preISSF1	0.0491	0.0285	0.0721	0.9014	0.8823
preISSF2	0.0455	0.0278	0.0696	0.9045	0.8880
preISSF3	0.0414	0.0264	0.0709	0.9046	0.8906
postISSF1	0.0447	0.0269	0.0683	0.9066	0.8900
postISSF2	0.0419	0.0257	0.0674	0.9086	0.8935
postISSF3	**0.0366**	**0.0212**	**0.0639**	**0.9161**	**0.9018**

$s' = 1$ is set for ARHS pansharpening methods, and the results of outputs are with the same resolution of the observed PAN image. Quality indicates D_s, D_λ, and QNR are used to make assessments on the results of different methods. As can be seen from Tables V and VI, preISSF methods and postISSF methods perform much better than other pansharpening methods, especially in spectral fidelity. In addition, our methods show high spatial fidelity and high spectral fidelity in the Merced's visualization of RGB images and false-color images in Fig. 19.

The above experimental results suggest the superiorities of our newly designed preISSF and postISSF for the ARHS pansharpening task. In addition, their effectivenesses on the standard pansharpening are also verified. It is noteworthy that preISSF1-preISSF3 achieve slightly different performances, which is also for postISSF1-postISSF3. This may be due to that they adopt different preliminary encoding processings. IC-DyPNN used in the encoding of preISSF3 and postISSF3

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TABLE VI
QUALITY METRICS OF FULL-RESOLUTION EXPERIMENT ON THE CHANGPING DATASET

Methods	$D_\lambda^F(\downarrow)$	$D_\lambda(\downarrow)$	$D_s(\downarrow)$	QNR(↑)	HQNR(↑)
Bicubic	0.1876	0.1921	0.2394	0.6144	0.6179
GSA	0.1796	0.1836	0.2503	0.6121	0.6151
SFIM	0.1221	0.1375	0.2480	0.6486	0.6602
MTF-GLP	0.1179	0.1264	0.2500	0.6551	0.6616
GFPCA	0.2022	0.2127	0.2375	0.6003	0.6083
BayesSparse	0.1178	0.1309	0.2553	0.6472	0.6570
PNN	0.1632	0.1739	0.0362	0.7962	0.8065
HyperPNN	0.1237	0.1574	0.0944	0.7630	0.7936
HSpeNet	0.0294	0.0301	0.0820	0.8903	0.8910
DSNet	0.0573	0.0679	0.0974	0.8413	0.8509
ARHSCNN	0.0254	0.0855	0.0580	0.8615	0.9181
VSPC-DARN	0.0478	0.0533	0.0352	0.9134	0.9187
preISSF1	0.0275	0.0301	0.0334	0.9375	0.9400
preISSF2	0.0262	0.0297	0.0317	0.9395	0.9429
preISSF3	0.0237	0.0267	**0.0281**	0.9459	0.9488
postISSF1	0.0234	0.0279	0.0383	0.9349	0.9391
postISSF2	0.0211	0.0232	0.0337	0.9438	0.9459
postISSF3	**0.0193**	**0.0197**	0.0319	**0.9480**	**0.9501**

employs a dynamic tactic which could adapt to the data nonstationary, and thus the two methods achieve relatively better performances. Meanwhile, postISSF tends to perform better than preISSF to some degree. This may be due to the two-stream INR structure design, which better accommodates the spectral–spatial characteristics of HS images.

VI. CONCLUSION

ARHS pansharpening is to use PAN images to enhance HS images to any desired spatial resolutions, which is more challenging than tradition standard HS pansharpening. To take on the ARHS pansharpening challenge, we propose two innovative TE-based ISSFs for ARHS pansharpening in this work, i.e., preISSF and postISSF. Our newly designs consider the cooperations of TE construction, spectral–spatial information integration, spectral–spatial query mapping, and coordinate normalization. In postISSF, the integration of spectral–spatial information occurs during the coordinate query mapping phase, which takes place after the TE feature extraction. In preISSF, TE feature extraction forms the spectral–spatial information integration before coordinate query mapping. In postISSF, the spectral–spatial information integration is completely in the duration of coordinate query mapping, which is after TE feature extraction. Especially, we design a novel mode-1 decomposition-based latent transformer to achieve effective global contextual information refinement, and develop novel explicit–implicit positional encoding systems to build strong connections of ensemble features and positions. Experimental results from several datasets suggest the superiorities of our two innovative ARHS pansharpening schemes. In the future, useful model considerations adopted in some classic pansharpening methods, such as the linear spectral mixing model and some valid optimization constraints, could be included in our ISSF designs.

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